

UNIT-1

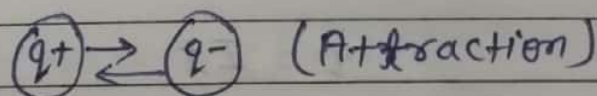
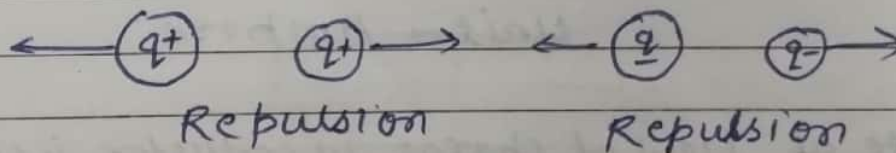
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DC CIRCUITS

- Electrical circuit Element (R, L, C).
- Voltage and current sources.
- Kirchhoff's current and voltage laws.
- Analysis of simple circuits with DC excitation.
- Superposition, Thevenin, Norton Theorem and maximum power transfer theorem
- star to delta conversion
- Time domain Analysis of first order RL and RC circuit.

* Charge :- It is an imaginary mechanism by which any substance experience a force of attraction or repulsion.



$$\text{Charge in } e^- = -1.6 \times 10^{-19} \text{ C}$$

$$\text{Charge in } p^+ = +1.6 \times 10^{-19} \text{ C}$$

$$\text{Mass of proton} = 1.67 \times 10^{-27} \text{ kg}$$

$$\boxed{\text{Mass of } e^- = \frac{\text{Mass of } p^+}{1840}}$$

* Number of charge = n

$$e^- = 1.6 \times 10^{-19} \text{ C}$$

$$q = ne$$

Q(1) Find total charge in an object where $n = 5$

$$e = 1.6 \times 10^{-19} \text{ C}$$

$$q = ne$$

$$= 5 \times 1.6 \times 10^{-19} \text{ C}$$

$$= 8 \times 10^{-19} \text{ C} \quad \text{Ans.}$$

Current :- It is a flow charge through a conductor. ~~it is~~
Unit - Ampere

Rate of flow of charge in conductor is called current

$$q = It$$

$$i = \frac{q}{t}$$

$$\text{If } t = 1 \text{ sec}$$

$$\text{then } i = q$$

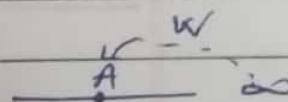
Q. Find current in a circuit in 10 sec
total charge in a conductor is 20 coulomb

Solⁿ, $t = 10 \text{ sec}$
 $q = 20 \text{ Coulomb}$

$$i = \frac{q}{t} = \frac{20}{10} = 2 \text{ Amp.}$$

* Potential - It is a workdone to bring a unit positive charge from infinity to at any point. S.I unit - volt

unit = Joule/coulomb



$$V = \frac{W}{q}$$

If, $q = 1 \text{ C}$

$$V = W$$

* E.M.F (Electro motive force) :-

It is a kind of pressure which is given to each coulomb of charge. Unit $\rightarrow$ volt

* Resistance :- It offer opposition to the flow of charge (current) in any circuit.

$$R = \frac{\rho L}{A}$$

Unit - ohm (Ω)

where, ρ = resistivity

A = cross sectional Area of conductor

L = Length of conductor.

* ρ depends on nature of material it is a fixed quantity.

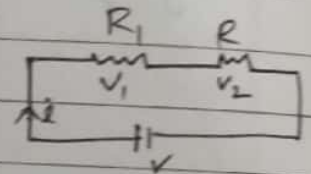
Resistance Combination:-

- (i) Series combination
- (ii) Parallel combination

(i) Series combination -

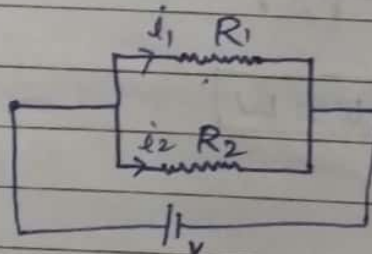


$$R_{eq} = R_1 + R_2$$



* current same in each resistor * voltage diff.

(ii) Parallel combination

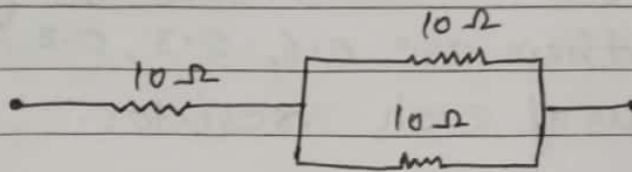


$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

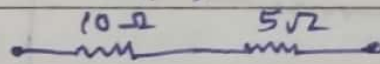
- current diff in each resistor
- voltage same.

Q. Find equivalent Resistance in following circuit.



$$R_{eq} = \frac{10 \times 10}{10 + 10}$$

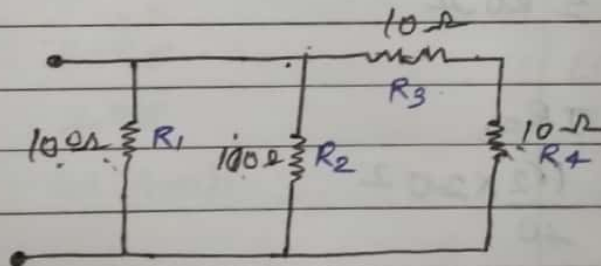
$$= \frac{100}{20} = 5 \Omega$$



$$R_{eq} = 10 \Omega + 5 \Omega$$

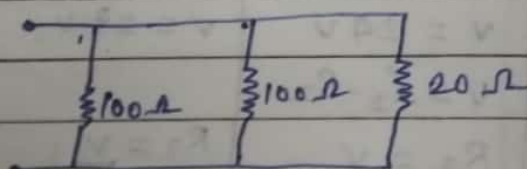
$$= 15 \Omega \text{ Ans.}$$

Q. Find $R_{eq} \rightarrow$



For, R_3 & R_4

$$R_A = 10 + 10 = 20 \Omega$$



$$R_{eq} = \frac{50 \times 20}{50 + 20}$$

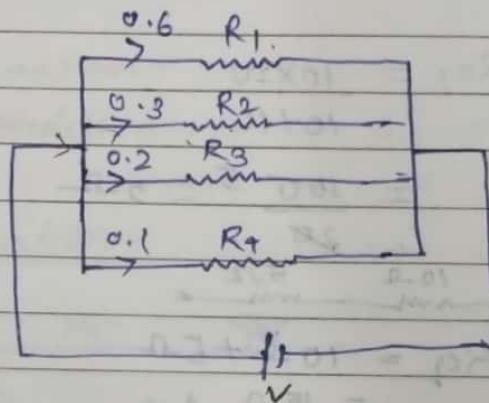
$$= \frac{1000}{70} = \frac{100}{7} \Omega$$

A_2

$$\frac{1}{R_{eq}} = \frac{1}{100} + \frac{1}{100} + \frac{1}{20}$$

$$\frac{1}{R_{eq}} = \frac{1}{50} + \frac{1}{20}$$

- Q. The equivalent Resistance of 4 Resistors joined in parallel is 20Ω . The currents flowing through them are 0.6 , 0.3 , 0.2 and 0.1 Amp. Find the value of each resistor.
Soln-



$$\text{Total current (i)} = 0.6 + 0.3 + 0.2 + 0.1 \\ = 1.2 \text{ Amp.}$$

$$R_{eq} = 20\Omega$$

$$V = IR_{eq} \\ = \frac{1.2 \times 20}{10} \\ = 24V$$

For $R_1 \rightarrow$

$$i_1 = 0.6$$

$$V = 24V$$

$$V = i_1 R_1$$

$$R_1 = \frac{V}{i_1}$$

$$= \frac{24 \times 40}{0.6}$$

$$R_1 = 40\Omega$$

For $R_2 \rightarrow$

$$i_2 = 0.3$$

$$V = 24V$$

$$V = i_2 R_2$$

$$R_2 = \frac{V}{i_2}$$

$$= \frac{80}{0.3}$$

$$R_2 = 80\Omega$$

For $R_3 \rightarrow$

$$i_3 = 0.2$$

$$V = 24V$$

$$R_3 = \frac{V}{i_3}$$

$$= \frac{120}{0.2}$$

$$R_3 = 120\Omega$$

For $R_4 \rightarrow$

$$i_4 = 0.1$$

$$V = 24V$$

$$R_4 = \frac{V}{i_4}$$

$$= \frac{240}{0.1}$$

$$R_4 = 240\Omega$$

Resistance.

- Q. Three loads A, B, C are connected in Parallel to a 240 volt source. Load A takes 9.6 kW, load B takes 60 Amp. and load C has a resistance of 4.8Ω calculate $\rightarrow R_A, R_B$, total current, total Power, Req.

Solⁿ-

For Load A —

$$P = 9.6 \text{ kW}$$

$$P = 9.6 \times 1000$$

$$P = 9600 \text{ watt}$$

we know that,

$$V = 240 \text{ V (Given)}$$

$$P = VI$$

$$I = \frac{P}{V}$$

$$= \frac{9600}{240}$$

$$= 40 \text{ Amp}$$

$$R_A = \frac{V}{I}$$

$$= \frac{240}{40} = 6 \Omega$$

$$R_A = 6 \Omega$$

For Load B $\rightarrow$

$$I = 60 \text{ Amp} \quad \left\{ \text{Given} \right\}$$

$$V = 240 \text{ volt}$$

$$P = VI$$

$$= 240 \times 60$$

$$= 14400 \text{ watt}$$

Note:-

(Watt)
Power = Voltage $\times$ Current
Electrical

$$P = VI$$

$$= (IR) I$$

$$P = I^2 R$$

$$P = \left(\frac{V}{R} \right)^2 \times R$$

$$P = \frac{V^2}{R}$$

Electrical

Energy = Power $\times$ Time

$$= VI \times t$$

$$= VI t$$

unit $\rightarrow$ Joule

$$E = I^2 R t$$

$$E = \frac{V^2}{R} t$$

$$1 \text{ unit} = 1 \text{ KWhr}$$

$$= 1000 \times 3600$$

$$= 3600000 \text{ Wsec}$$

$$1 \text{ unit} = 3.6 \times 10^6 \text{ wattsec}$$

$$R_B = \frac{V}{i}$$

$$= \frac{240}{60} = 4 \Omega$$

Load C -

$$R = 48 \Omega$$

$$V = 240 V$$

$$V = iR$$

$$i = \frac{V}{R}$$

$$i = \frac{240}{48} = 50 \text{ Amp.}$$

$$P = Vi$$

$$P = 240 \times 50$$

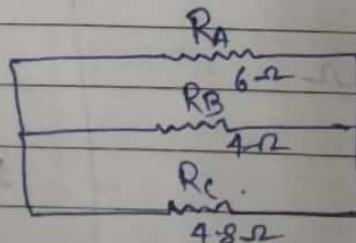
$$= 12000$$

$$\text{Total current} \rightarrow 40 + 60 + 50$$

$$= 150 \text{ Amp.}$$

$$\text{Total Power} = 9600 + 14400 + 12000$$

$$= 36000 \text{ watt}$$



$$\frac{1}{R_{eq}} = \frac{1}{6} + \frac{1}{4} + \frac{1}{48}$$

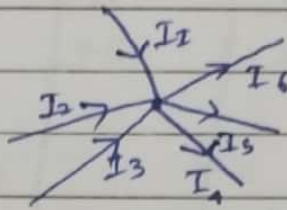
$$= \frac{1}{6} + \frac{1}{4} + \frac{5}{24} = \frac{4+6+5}{24} = \frac{15}{24}$$

$$R_{eq} = \frac{24}{15} \Omega \text{ Ans}$$

Kirchhoff laws →

Kirchhoff's current law (KCL)

According to this law at a meeting point incoming current and outgoing current algebraic summation is 0.

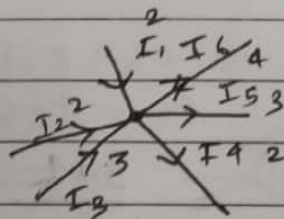


$$I_1 + I_2 + I_3 - (I_4 + I_5 + I_6) = 0$$

$$I_1 + I_2 + I_3 - I_4 - I_5 - I_6 = 0$$

$$\boxed{\sum I = 0}$$

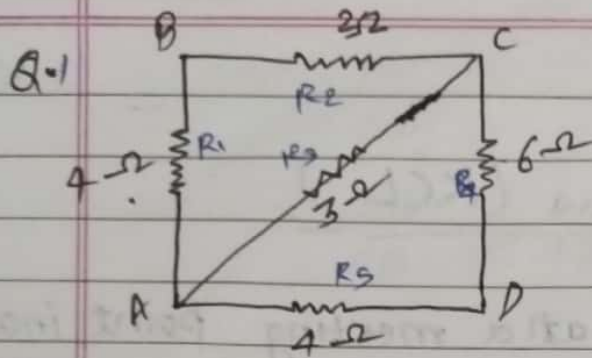
Q.



$$\Rightarrow 2 + 5 + 1 - (5 + 3 + 2) = 0$$

$$8 - 8 = 0$$

$$\underline{0 = 0} \quad \#$$



solnⁿ ①

$R_1 + R_2$ are in series

$$R_A = 4 + 2 = 6 \Omega$$

$R_A + R_3$ are in parallel

$$\frac{1}{R_B} = \frac{1}{6} + \frac{1}{3} = \frac{1+2}{6} = \frac{3}{6} = \frac{1}{2} \Rightarrow R_B = 2 \Omega$$

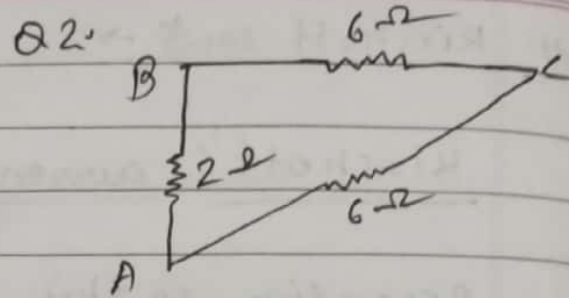
$$R_B = 2 \Omega$$

$R_4 + R_5$ are in series

$$R_C = 6 + 4 = 10 \Omega$$

$$\frac{1}{R_{eq}} = \frac{1}{2} + \frac{1}{10} = \frac{5+1}{10} = \frac{6}{10} = \frac{3}{5} \Rightarrow R_{eq} = \frac{5}{3} \Omega$$

$$R_{eq} = \frac{5}{3} \Omega$$

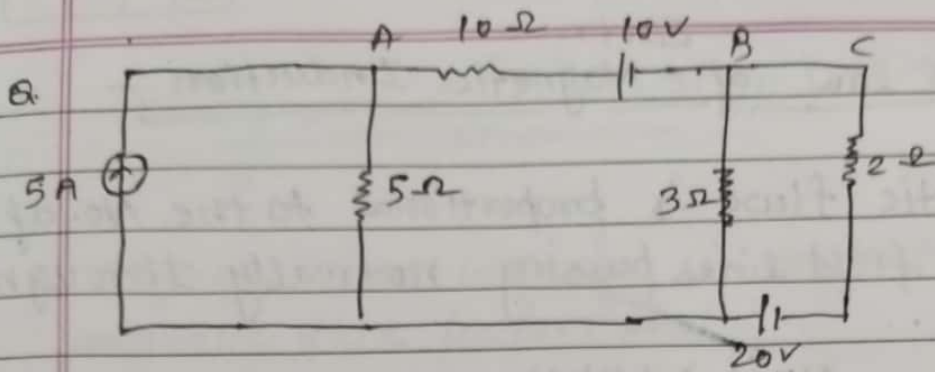


$R_1 + R_2$ are in series

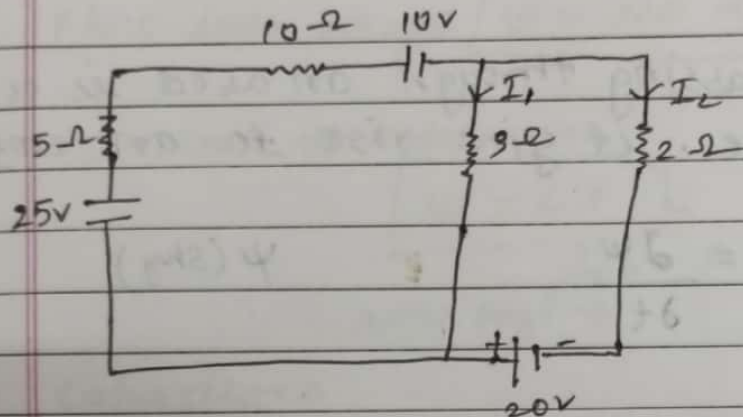
$$R_A = 6 + 2 = 8 \Omega$$

$$\frac{1}{R_T} = \frac{1}{8} + \frac{1}{6} = \frac{3+4}{24} = \frac{7}{24} \Rightarrow R_T = \frac{24}{7} \Omega$$

$$\begin{array}{r} 2 \overline{) 3.6} \\ \underline{2 3} \\ 2 3 \\ \underline{2 3} \\ 0 \end{array}$$



$$\begin{aligned} 2\Omega &\rightarrow (1) \\ 4\Omega &\rightarrow (2) \\ 5\Omega &\rightarrow (3) \\ 2\Omega &\rightarrow (4) \end{aligned}$$



$$(5+10)I + 3I_1 = 25-10$$

$$15I + 3I_1 = 15 \quad \text{--- (i)}$$

$$2I_2 - 3I_1 = 20 \quad \text{--- (ii)}$$

$$I = I_1 + I_2 \quad \text{--- (iii)}$$

$$15(I_1 + I_2) + 3I_1 = 15$$

$$15I_1 + 15I_2 + 3I_1 = 15$$

$$18I_1 + 15I_2 = 15 \quad \text{--- (iv)}$$

$$3(6I_1 + 5I_2) = 15$$

$$6I_1 + 5I_2 = 5 \quad \text{--- (v)}$$

$$\text{eqn (ii)} \times 2$$

$$- (6I_1 + 4I_2) = 40$$

$$-6I_1 + 4I_2 = 40$$

$$6I_1 + 5I_2 = 5$$

$$9I_2 = 45$$

find the voltage drop
across 2Ω —

$$\begin{aligned} V &= 2 \times 5 \\ &= 10V \end{aligned}$$

$$I_2 = \frac{45}{9}$$

$$I_2 = 5 \text{ Amp.}$$

value of I_2 put in eqn (v)

$$6I_1 + 25 = 5$$

$$6I_1 = -20$$

$$I_1 = \frac{-20}{6} = -\frac{10}{3} \text{ Amp.}$$

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 Electro Magnetic Induction :-
Faraday's law of

A magnetic flux is proportional to the No. of magnetic field lines passing normally through an area.

unit $\rightarrow$ Weber

If the flux passing through an area is a function of time, it give rise to an emf.

$$e = \frac{d\psi}{dt}$$

ψ (sky)

$$\psi = N \cdot \phi$$

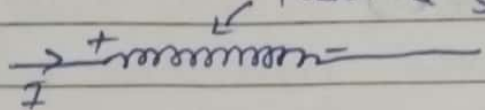
$$e = \frac{d}{dt} (N\phi)$$

$$e = N \frac{d\phi}{dt}$$

If the flux vary with time then induced e.m.f.

Self Inductance :-

If the flux produced by a coil is responsible for induced emf with in the same coil. this process is called self inductance.

Passive sign convention.


Magnetic field intensity is proportional to I then magnetic flux proportional I .

Flux into linkage $\boxed{\psi = N\phi \propto I}$

In case of inductance-

$$\boxed{\psi = L i}$$

Next page $\rightarrow$ P.T.O $\rightarrow$

capacitors

capacitor is the property of a circuit element or device which is capable of storing charge or energy in electric field. A two terminal circuit element is called a capacitor, if at any time (t) it is stored energy $q(t)$ and voltage $V(t)$ across it are related algebraic relation $q(t)$ into $V(t)$ equal to zero.

$$\int [q(t) \cdot v(t)] = 0$$

With time invariant

$$\boxed{q(t) = C \cdot V_c(t)}$$

current through the capacitor can be obtain from the relation.

$$i(t) = \frac{d}{dt} q(t)$$

$$i_c(t) = \frac{d}{dt} C v_c(t)$$

$$i_c(t) = C \cdot \frac{dv_c}{dt}$$

$$\int_{t_0}^t dv_c = \int \frac{1}{C} \cdot i_c(t) dt$$

Energy stored in a capacitor:-

Capacitors are energy storing elements. The energy stored in a time invariant capacitor at time (t) .

$$W_c(t) = \frac{1}{2} C v_c^2(t) \quad \text{** Proof:-}$$

$$W_c = \int_{-\infty}^t v_c(t) \cdot i_c(t) dt$$

KVL $\Rightarrow$ The ~~the~~ algebraic sum of all voltages around a closed path at any instant is zero.

$$\sum V(t) = 0 \quad \text{**}$$

The closed path is to be traced in any arbitrary specified direction which may be clockwise & anticlockwise, in most of the cases ~~we~~ we shall choose clockwise direction for the closed path.

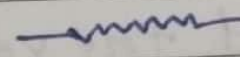
The voltage across element that are result ~~traversed~~ traversed -ve to +ve are taken +ve and the voltage across element that are ~~result~~ travel from +ve to -ve are taken -ve.

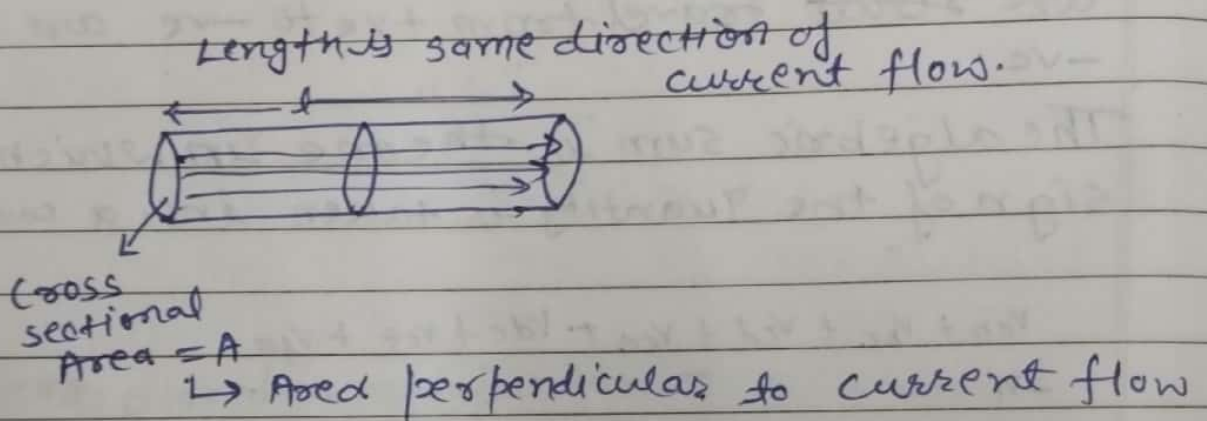
The algebraic sum is the one in which the sign of the quantity is taken into account.

$$V_{ab} + V_{bc} + V_{cd} + V_{ce} + V_{de} + V_{ef} + V_{fa} \\ -V_1$$

* Resistor :- A Resistor is an element that opposes the flow of charge or current through it. That is called is Resistor.

Resistance is the property of an element by which it opposes flow of current.

S.I unit $\rightarrow$ ohm (Ω), 

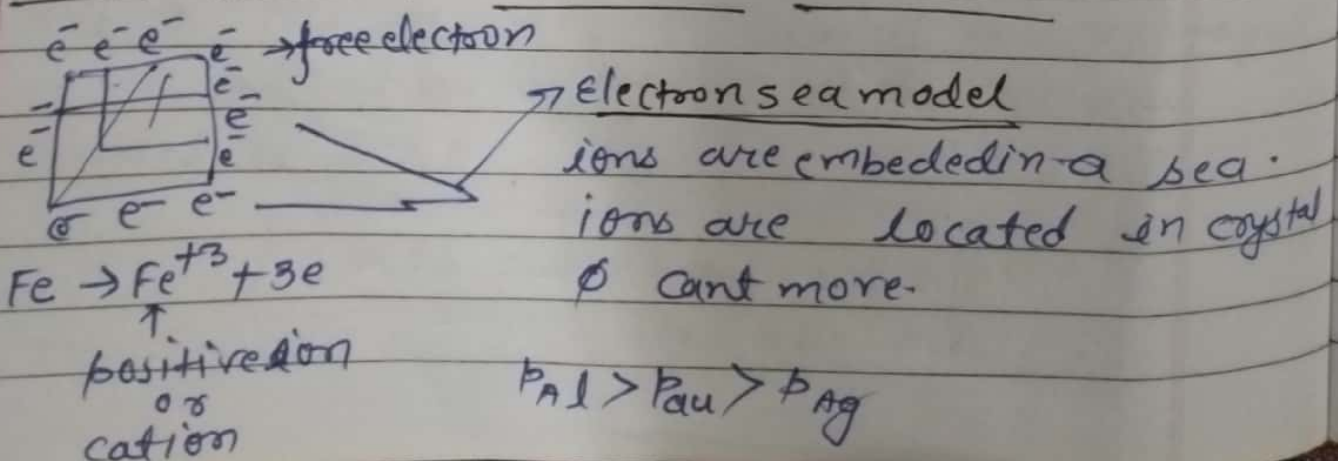


Resistance, $R = \frac{\rho \cdot l}{A} \Omega$

ρ = resistivity

Property of a material. (Opposition to current flow it does not depend on dimension it is depend only materials. S.I unit. ($\Omega \cdot m$).

* Why any material opposes the current :-

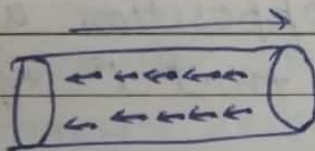


Electron motion :-

Electron motion of e^-

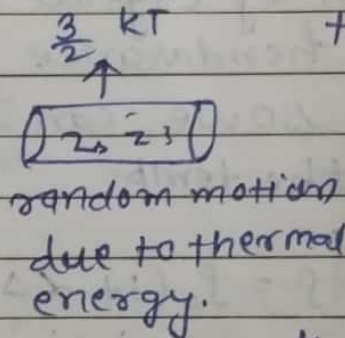
drift velocity
↓

due to force of electric field



Thermal velocity when we are placed any material on room temp. than create a thermal energy.

↓
random motion due to temp.



$K = \text{Boltzmann constant}$

$$1.38 \times 10^{-23} \text{ J/K}$$

$T = \text{temp.}$

Due to random motion e^- collide among themselves as well as with ions leading to hindrance in the motion of e^- producing resistance.

Super Conductor :- At 0°K temp. the thermal energy of ions, as well as electrons, becomes zero so there will be no vibration or random motion and hence there are no collisions between electrons or b/w

blw electrons and ions, so the resistance. Become zero. these types of conductors called super conductor.

Temperature coefficient :-

When temp. increases than thermal energy of Both types of ions (electrons & ions) and randomness in e^- and ions are increased, so vibration of crystal ions is increased. there for hindrance or opposition are increased so we can say that R of metal $\uparrow$ n with the temp.

$$\rho = \rho_0 (1 + \alpha \Delta T)$$

ρ_0 = resistivity at temp T_0

ρ = resistivity at temp T

$$\Delta T = T - T_0$$

α = temperature coefficient (% or $^{\circ}\text{K}^{-1}$)

$$R = R_0 (1 + \alpha \Delta T)$$

→ same relations hold true for R .

ohm's law -

When a potential difference is applied across the ends of a conductor a current start to flow b/w +ve two ends

proportional to potential difference applied across ends of conductor at constant temp.

$$I \propto V$$

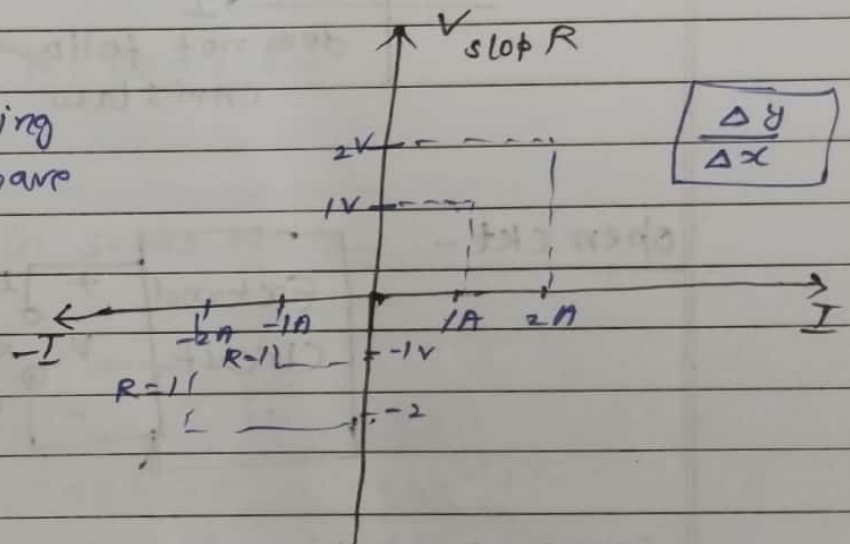
$$I = \frac{V}{R} \quad (\text{Resistance})$$

$$\boxed{V = IR}$$

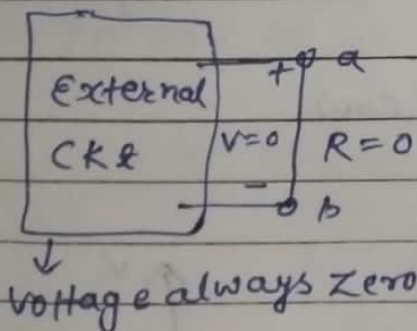
V-I Characteristics of Linear Resistor:-

$$V = IR$$

→ straight line passing through origin have a slope resistance.

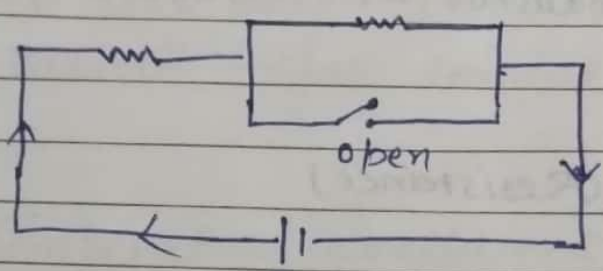


Concept of Short Circuit:-

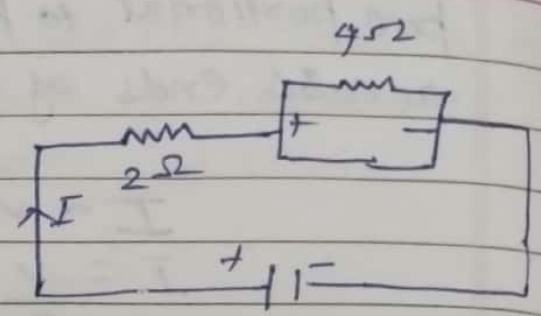


When 2 nodes are connected by wire or a perfect conductor is known as short circuit. Then $R=0$, then both end point potential is equal. $V_a = V_b = 0$. difference. current can have any value its depends.

on ckt open ckt

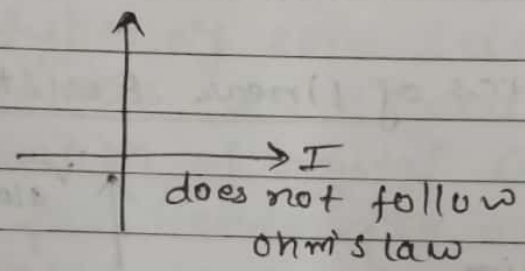


$$I = \frac{12}{2+4} = 2 \text{ Amp.}$$

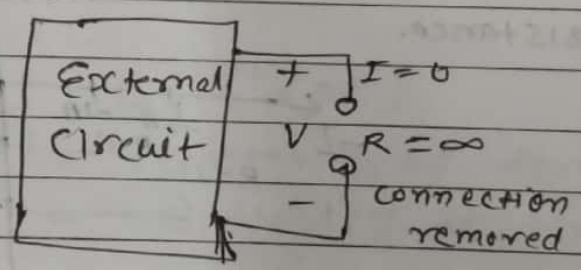


Enter current through short ckt.

$$I = \frac{12}{2} = 6 \text{ Amp.}$$



open ckt -



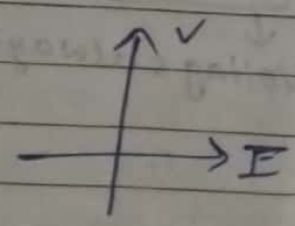
Since connection is not available b/w 2 nodes current flow $I = 0$.

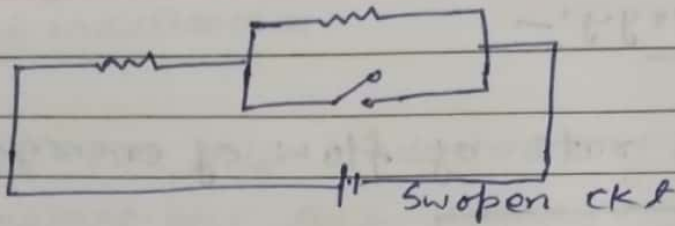
voltage b/w two nodes can have depends on external ckt.

If we are Apply ohms law.

$$R = \frac{V}{I} = \infty$$

Because $R = \frac{V}{0}$





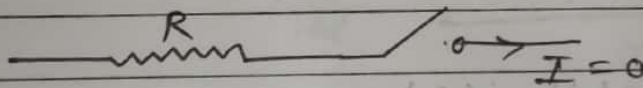
$$I = \frac{12}{u+2} = 2A$$

$$V = V_{u-2} = I \times u - 2$$

open ckt and short ckt. in case of resistance.

In case of R, ohm's law is valid.

for open ckt in series with

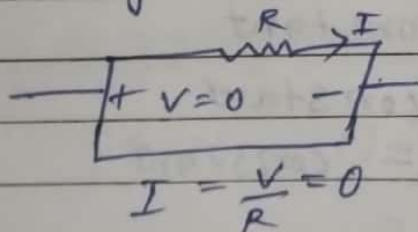


$$V = I \cdot R = 0 \times R = 0$$

Both V and I = 0

In case of short ckt.

In case of short ckt. across R



$$I = \frac{V}{R} = 0$$

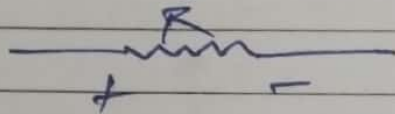
Both V and I are zero. In Resistance cases open ckt and short ckt both are zero.

* Power and energy:-

power is the rate of flow of energy.

$$P = V \cdot I \quad \text{watt.}$$

In case of resistance.



$$V(t) = i(t) \cdot R$$

Resistance absorbs power

Resistance convert electrical energy into heat.

Instantaneous power:-

$$P(t) = V(t) \cdot i(t)$$

$$V(t) = [i(t)]^2 \cdot R$$

$$i(t) = \frac{[V(t)]^2}{R}$$

if we are used d.c. supply

$$i(t) = I = \text{constant}$$

$$V(t) = V = \text{constant}$$

$$I^2 R = V^2 / R = \text{constant}$$

$$\text{Energy} = \int P(t) \cdot dt$$

if d.c. supply then energy $I^2 R t = \frac{V^2}{R^2}$

* Self inductance *

L is the self inductance of a coil proportionality constant b/w flux linkage + current.

unit:- Henry (H)

$$\text{emf} = d\psi/dt = \frac{d}{dt}(Li)$$

$$e = L \frac{di}{dt} = v$$

power & energy'

$\rightarrow i$ mmmmm

$$v = L \frac{di}{dt}$$

$\frac{di}{dt}$; rate of change of current

$$\text{power observed} = v(t) \cdot i(t)$$

$$= L \frac{di(t)}{dt} \cdot i(t)$$

$$= L i(t) \cdot \frac{di(t)}{dt} \rightarrow \text{instantaneous power}$$

Energy is getting absorb:-

$$\int p(t) \cdot dt = \int L i \, di$$

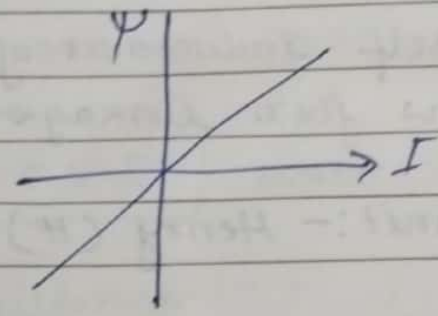
$$= \frac{1}{2} LI^2$$

energy Absorbed b/w $t = t_1$ & $t = t_2$ $= \frac{1}{2} Li^2$

$$\Delta u = u(t_2) - u(t_1)$$

$$= \frac{1}{2} L [i^2(t_2) - i^2(t_1)]$$

ψ - I characteristics :-



linear magnetic material
 $\psi \propto I \rightarrow$ paramagnetic materials.

$L = \text{constant}$
 When $\psi \propto I$

Ferromagnetic material have non linear behaviour
 But we ignore non linearity.

Inductor opposes change of current

Len's law - According to this law effect
 the opposes cause that is
 when we are change due current in inductor
 change the magnetic field.

A/c to this

Induced e.m.f opposes the cause of induction

$$e = L \frac{di}{dt}$$

cause of emf change of current
 with respect to time

* Induced emf tends to maintain current constant inductor allows the flow of constant current But opposes the change of current

$$\text{If } i(t) = \text{constant}$$

$$\frac{di}{dt} = 0$$

emf will be zero

no. emf induced then no oppos

$$e = 0 \rightarrow \text{no opposition}$$

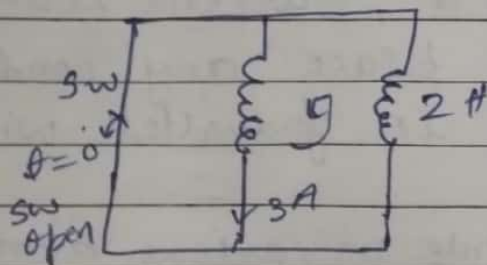
$\rightarrow$ short CK

$i = 0$ than open CK

* In inductor current not change suddenly.

Generally we say that current through an inductor can not change instantly as emf tends to ∞

But due to magnetic inertia flux linkage should not change instantly. But in some cases current can change instantly.

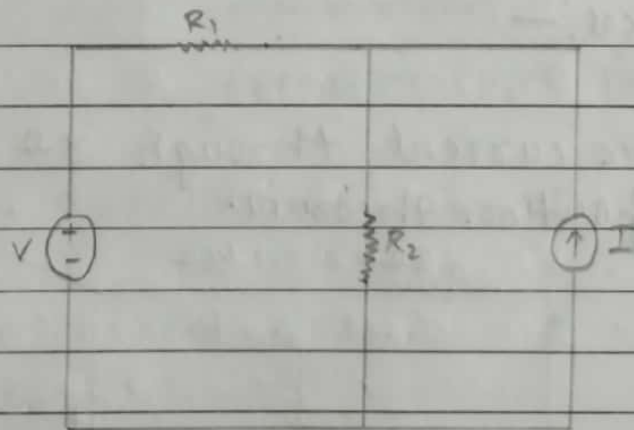


Superposition theorem :-

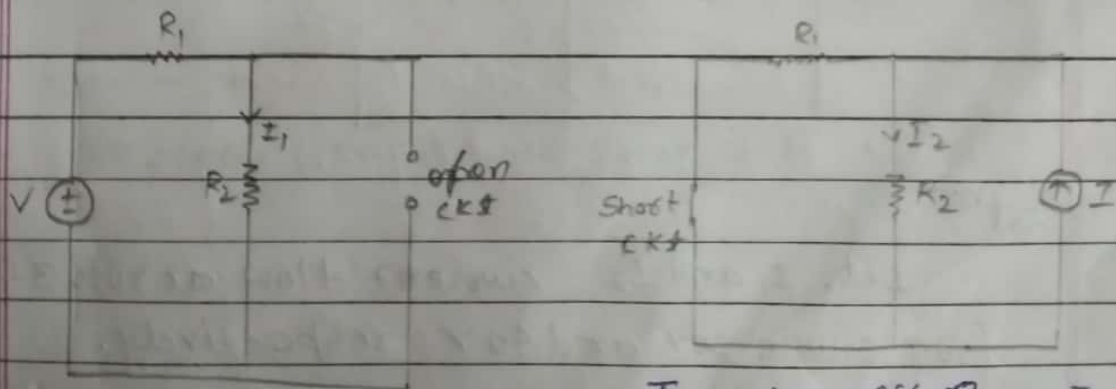
In any linear network containing resistors and other components with more than one source of voltages (or current sources).

The response (voltage and current) in any element is equal to the sum of all responses caused by individual source acting alone.

While other source are eliminated from the circuit. And voltage source replaced by the short ckt and current source replaced by the open circuit.



Take one source at a time.



$$I_{\text{total across } R_2} = I_1 + I_2$$

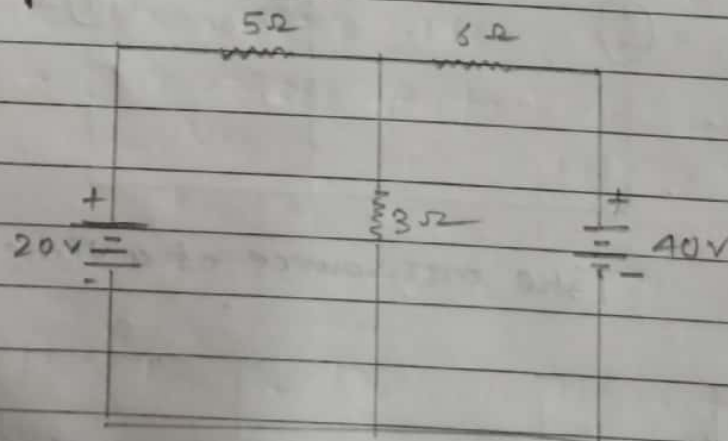
$$V_{\text{total across } R_2} = V_1 + V_2$$

The Drawbacks:-

- (i) Superposition theorem cannot apply to the circuit having only a dependent source. It needed atleast two independent source.
- (ii) This theorem only applicable to the network which consists of linear elements. It cannot apply to the non-linear elements like a diode, transistor etc.
- (iii) We cannot calculate power by this theorem. As the power is proportional to the square of voltage and current as it becomes non-linear.

Numerical:-

Find the current through 3Ω resistor using superposition theorem.



Solⁿ-

Let, I_1 and I_2 current flow across 3Ω resistor from source 20V and 40V respectively.

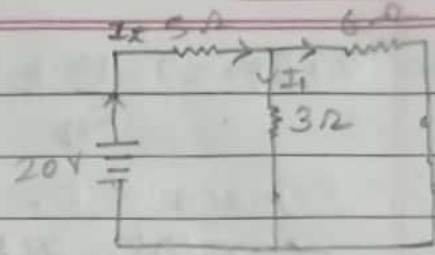
Replacing 40V source by short ckt.

$$R_{eq} = 5 + 3 \parallel 6$$

$$= 5 + \frac{3 \times 6}{6 + 3}$$

$$= 5 + \frac{18}{9}$$

$$R_{eq} = 7 \Omega$$



$$I_T = \frac{V}{R} = \frac{20}{7} = 2.857 \text{ Amp.}$$

current across 3Ω resistor

Applying current division rule

$$I_1 = I_T \times \frac{\text{oppo. resistance}}{\text{oppo. resistance} + \text{current resistance}}$$

$$= 2.857 \times \frac{6}{3 + 6}$$

$$= 2.857 \times \frac{6}{9}$$

$$= 2.857 \times 0.66$$

$$I_1 = 1.885 \text{ amp.}$$

Now —

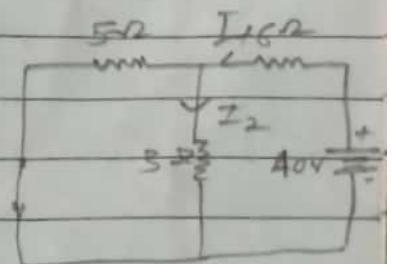
Replacing 20V source short CK &

$$R_{eq} = 6 + 3 \parallel 5$$

$$= 6 + \frac{3 \times 5}{3 + 5}$$

$$= 6 + \frac{15}{8}$$

$$R_{eq} = \frac{63}{8} = 7.875 \Omega$$



$$I_T = \frac{V}{R_{eq}} = \frac{40}{7.87} = 5.082 \text{ Amp.}$$

current across 3Ω resistor
current division rule. —

$$I_2 = I_T \times \frac{\text{oppo. resistance}}{\text{Total resistance}}$$

$$= I_T \times \frac{5}{3+5}$$

$$= 5.082 \times \frac{5}{8}$$

$$= 5.082 \times 0.625$$

$$I_2 = 3.176 \text{ Amp}$$

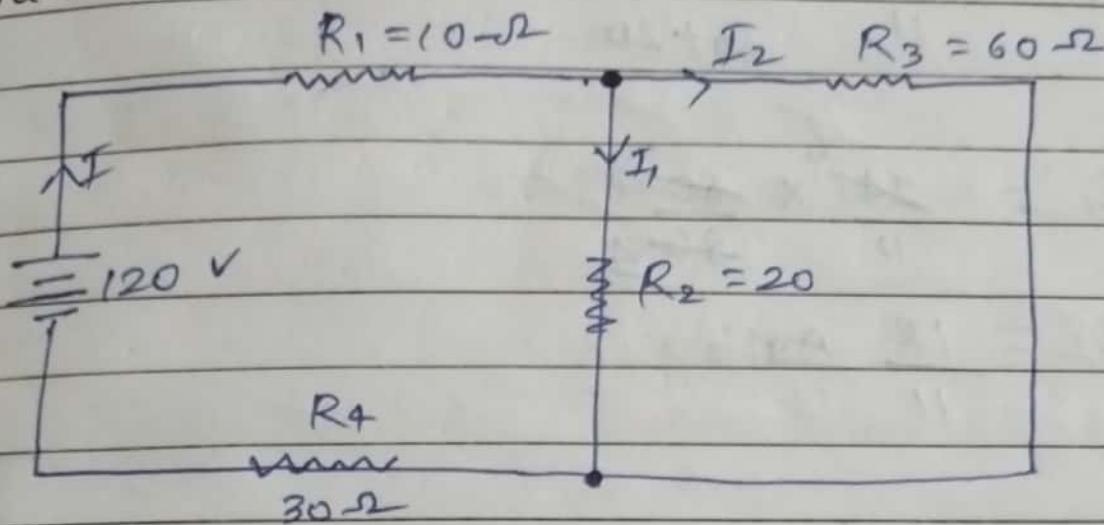
Total current across R_3 resistor

$$I = I_1 + I_2$$

$$= 1.885 + 3.176$$

$$I = 5.061 \text{ Amp.}$$

Numerical:-



By means of superposition theorem find the current which flows through R_2 in the circuit of.

$$\begin{aligned} R_{eq} &= R_1 + R_4 + R_2 \parallel R_3 \\ &= 10 + 30 + 20 \parallel 60 \\ &= 10 + 30 + \frac{20 \times 60}{80} \end{aligned}$$

$$= 10 + 30 + 15$$

$$R_{eq} = 55\Omega$$

$$V = 120 \text{ volt (given)}$$

$$I_T = \frac{120}{55} \text{ Amp} = \frac{24}{11} \text{ Amp}$$

from current division rule-

$$I_1 = I_T \times \frac{\text{Resistance of opp. Branch}}{\text{Resistance of opp. Branch} + \text{Resistance of current Branch}}$$

$$I_1 = \frac{24}{11} \times \frac{60}{60+20}$$

$$I_1 = \frac{24}{11} \times \frac{60}{80}$$

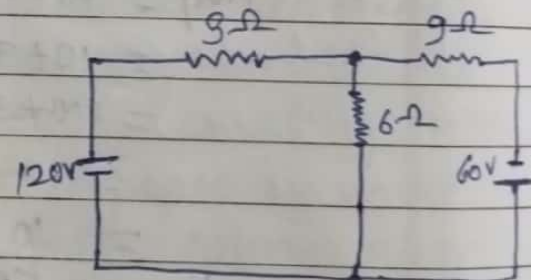
$$I_1 = \frac{18}{11} \text{ Amp}$$

$$I_1 = 1.636 \text{ Amp. as,}$$

Q2 → In the circuit show in fig. find the current through the 6Ω resistor using superposition theorem:-

solⁿ-

Replacing $60V$ source by a short circuit then fig.

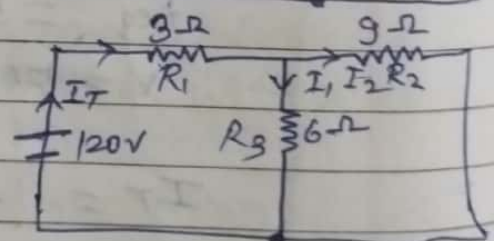


$$R_{eq} = 3 + 6 \parallel 9$$

$$= 3 + \frac{6 \times 9}{6+9}$$

$$= 3 + \frac{54}{15}$$

$$R_{eq} = \frac{54}{15} = 3.6 \quad \frac{45+54}{15} = \frac{99}{15} = 6.64 \Omega$$



Thevenin

$$I_T = \frac{120}{6+6} \times 10 = \frac{200}{11} \text{ Amp}$$

current in resistance R_3 —

using current division rule —

$$I_1 = I_T \times \frac{\text{Resis. of opp Branch}}{\text{Resis. of opp Branch} + \text{Resistance of current Branch}}$$

$$I_1 = \frac{200}{11} \times \frac{9}{9+6}$$

$$I_1 = \frac{40}{200} \times \frac{9}{158}$$

$$I_1 = \frac{120}{11} \text{ Amp}$$

similarly on replacement of 120V source by a short circuit —

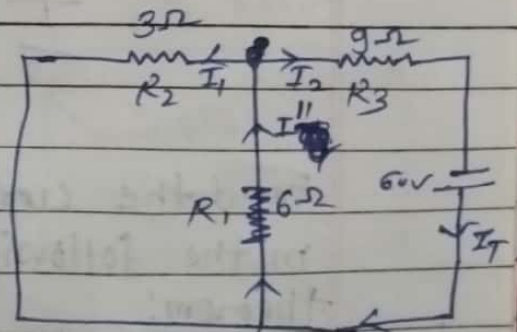
$$R_{eq} = 9 + 6 \parallel 3$$

$$= 9 + \frac{6 \times 3}{6+3}$$

$$= 9 + \frac{18}{9}$$

$$= 9 + 2$$

$$R_{eq} = 11 \Omega$$



$$I_{q_1} = \frac{60}{11} \text{ Amp}$$

current in resistance $R_1 \rightarrow$

using current division rule $\rightarrow$

$$I'' = \frac{60}{11} \times \frac{3}{3+6}$$

$$= \frac{20}{11} \times \frac{3}{3}$$

$$I'' = \frac{20}{11} \text{ Amp}$$

Total current through 6Ω resistor is equal to the algebraic sum of I_1 & I'' current -

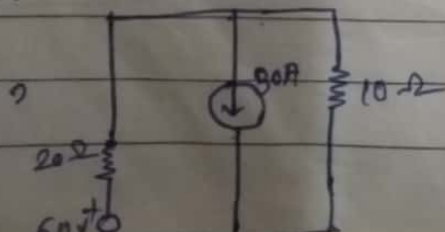
$$I = I_1 + I''$$

$$= \frac{120}{11} + \left(-\frac{20}{11} \right) \quad \left\{ \because I'' \text{ current is opposite direction of } I_1 \right\}$$

$$= \frac{120}{11} - \frac{20}{11}$$

$$I = \frac{100}{11} \text{ Amp Ans}$$

Find the current flowing through 10Ω resistance in the following circuit (Fig. 2.38) use Superposition theorem.



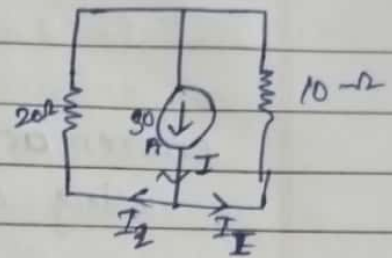
Replacing 60V source from short circuit

using current division rule -

$$I_1 = 30 \times \frac{20}{20+10}$$

$$= 30 \times \frac{20}{30}$$

$$I_1 = 20 \text{ Amp (upward)}$$

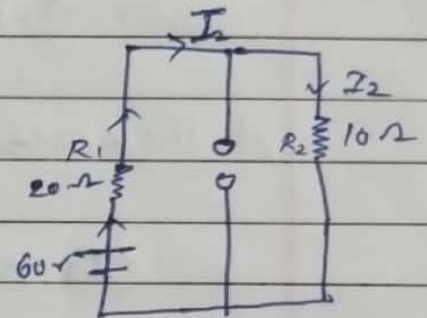


similarly Replacing 30 Amp current source by open circuit

current in resistance $R_2 \rightarrow$

$$I_2 = \frac{60}{30} = 2$$

$$I_2 = 2 \text{ Amp (downward)}$$



$$\begin{aligned} \text{Total current in resistance } 10\Omega &= 20 - 2 \\ &= 18 \text{ Amp} \end{aligned}$$

* By super position theorem Find the current in R.

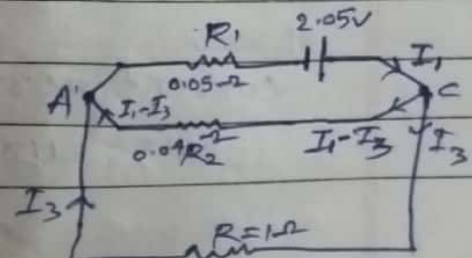
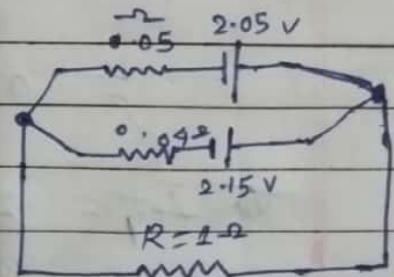
Consider $E_2 = 0$

$$R_{eq} = R_1 + R_2 \parallel R_3$$

$$= 0.05 + \frac{0.04 \times 1}{0.04 + 1}$$

$$= 0.05 + \frac{0.04}{1.04}$$

$$R_{eq} = 0.0885 \Omega$$



$$\text{Total current } (I_T) = \frac{2.05}{0.0885}$$

$$= 23.2 \text{ Amp.}$$

Current across resistance R —
using current division rule—

$$I_3 = I_T \times \frac{\text{Resistance in oppo. branch}}{\text{Resistance in oppo. branch} + \text{Resistance in current branch}}$$

$$= 23.2 \times \frac{0.04}{0.04 + 1}$$

$$= 23.2 \times \frac{0.04}{1.04}$$

$$I_3 = 0.892 \text{ Amp.}$$

NOW, $E_1 = 0$,

$$R_{eq} = R_2 + R_1 \parallel R_1$$

$$= 0.04 + \frac{1 \times 0.05}{1 + 0.05}$$

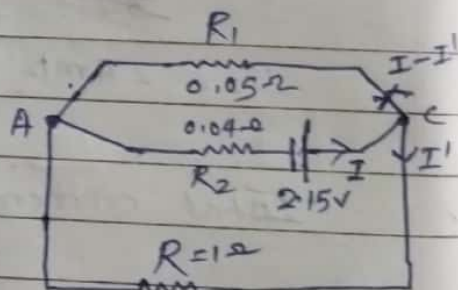
$$= 0.04 + \frac{0.05}{1.05}$$

$$= 0.876 \Omega$$

$$I_T = \frac{2.15}{0.876} = 2.454 \text{ Amp}$$

current across R →

using current division rule—



$$I' = I_T \times \frac{\text{Resis of opp branch}}{\text{Resis of opp branch} + \text{Resis of current branch}}$$

$$I' = 24.54 \times \frac{0.05}{0.05 + 1}$$

$$I' = \frac{24.54 \times 0.05}{1.05}$$

$$I' = 1.169 \text{ Amp}$$

Total current through 12 resistance

$$I = I' + I_3$$

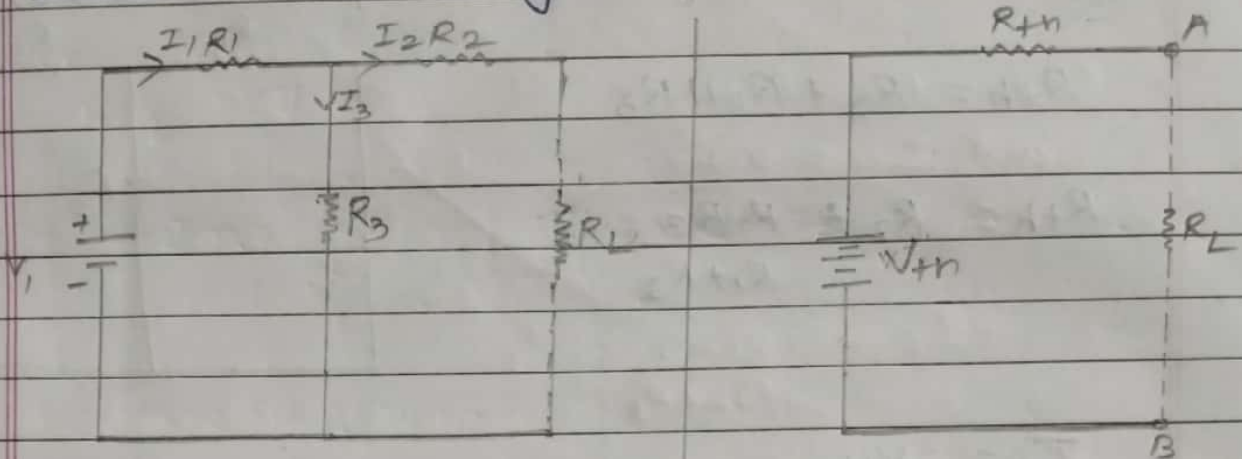
$$= 0.892 + 1.169$$

$$= 2.061 \text{ AMP}$$

Current division rule :-

$$I' = \text{Total current} \times \frac{\text{Resis of opp branch}}{\text{Resis of opp branch} + \text{Resistance of current branch}}$$

Thevenin theorem:- Any linear circuit containing several voltages and resistances can be replaced by just one ~~any~~ voltage (V_{th}) in series with a single resistance (R_{th}).



Step 1:- Remove R_L (i.e. terminals A and B are open circuited)

The current in R_1 and R_3

$$I_1 = \frac{V}{R_1 + R_3}$$

voltage across R_3 is

$$V_p = I_1 R_3$$

$$V_p = \frac{V R_3}{R_1 + R_3}$$

As terminal A and B are open circuited, there is no current in R_2 and the P.d V_{th} across A and B is:-

$$V_{th} = V_p = \frac{V R_3}{R_1 + R_3}$$

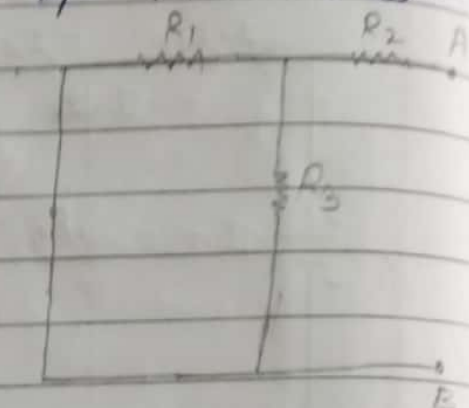
Step 2:- To find Thevenin resistance R_{th}

Replace V by short circuit

The resistance of network b/w A and B is -

$$R_{th} = R_2 + R_1 \parallel R_3$$

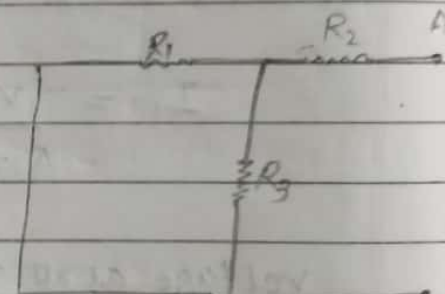
$$R_{th} = R_2 + \frac{R_1 R_3}{R_1 + R_3}$$



$$I_2 = \frac{V_{th}}{R_{th}}$$

$$I_2 = \frac{V R_3 / (R_1 + R_3)}{R_2 + \left(\frac{R_1 R_3}{R_1 + R_3} \right) + R_L}$$

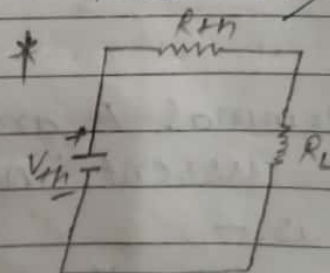
~~$$I_2 = \frac{V R_3}{R_{th} + R_L}$$~~



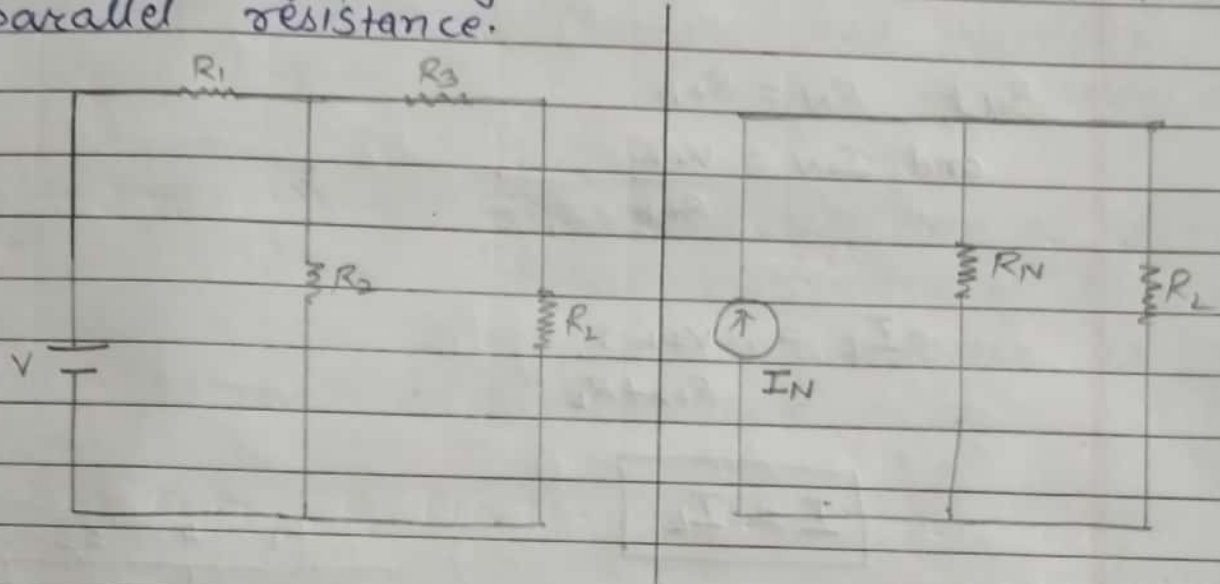
$I_2 = \frac{V_{th}}{R_{th} + R_L}$

At last R_2 Resistance connected^B again b/w terminal A and B.

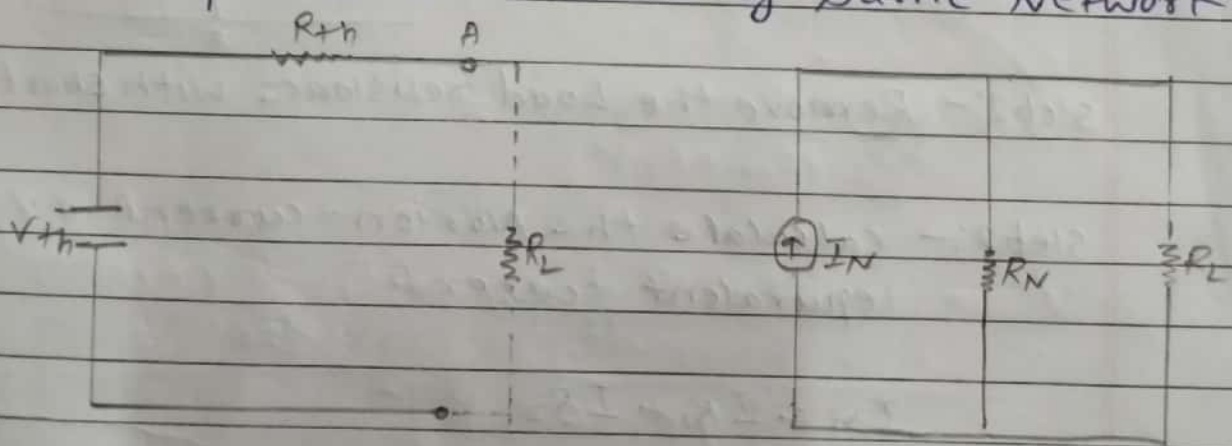
This proves thevenin theorem.



Norton theorem:- Any linear circuit containing several voltages and resistances can be replaced by an equivalent circuit consisting of a single current source and parallel resistance.



Proof:- consider the thevenin and Norton equivalent circuits of same Network.



The current I for Thevenin equivalent circuit in load resistance R_L

$$I = \frac{V_{th}}{R_{th} + R_L}$$

Now the current I_L for Norton circuit

equivalent circuit in load R_L

$$I_L = \frac{I_N R_N}{R_N + R_L}$$

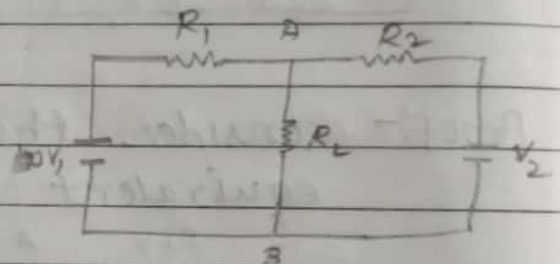
But, $R_N = R_{th}$

$$\text{and } I_N = \frac{V_{th}}{R_{th}}$$

$$I_L = \frac{V_{th}}{R_{th} + R_L}$$

$$I = I_L$$

Proved.



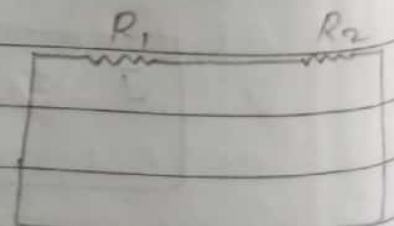
Step 1:- Remove the Load resistance with short circuit

Step 2:- calculate the Norton current or equivalent current

$$I_N = I_{R_1} + I_{R_2}$$

$$I_N = I(R_1 + R_2)$$

Step 3:- Replace the power source with short circuit



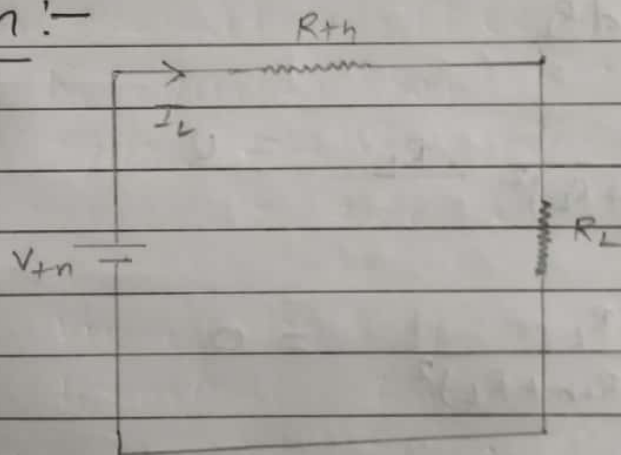
Step 4:- calculate Norton Resistance or equivalent resistance

Maximum power transfer theorem:-

Maximum power transfer theorem states that the maximum power is transferred to the load when the load resistance is equal to the thevenin resistance (R_{th}).

Thevenin resistance:- Thevenin resistance is the total resistance of the circuit when the load resistance is not present in the circuit.

Derivation:-



Power transfer to the load -

$$P_L = (I_L)^2 \cdot R_L \quad \text{--- (i)}$$

We know that
from thevenin theorem -

$$I_L = \frac{V_{th}}{R_{th} + R_L}$$

substituting the value of I_L in eqn (i) we get

$$P_L = \left(\frac{V_{th}}{R_{th} + R_L} \right)^2 \cdot R_L$$

$$P_L = \frac{V_{th}^2}{(R_{th} + R_L)^2} \cdot R_L \quad \text{--- (ii)}$$

For maximum of P_L

$$\frac{dP_L}{dR_L} = 0$$

$$\frac{d}{dR_L} \frac{V_{th}^2}{(R_{th} + R_L)^2} \cdot R_L = 0$$

$$V_{th}^2 \frac{d}{dR_L} \frac{R_L}{(R_{th} + R_L)^2} = 0$$

$$V_{th}^2 \cdot \left[\frac{(R_{th} + R_L)^2 \frac{d(R_L)}{dR_L} - R_L \frac{d(R_{th} + R_L)^2}{dR_L}}{[(R_{th} + R_L)^2]^2} \right] = 0$$

$$\frac{V_{th}^2 [(R_{th} + R_L)^2 \cdot 1 - R_L \cdot 2(R_{th} + R_L) \cdot 1]}{(R_{th} + R_L)^4} = 0$$

$$\frac{V_{th}^2 (R_{th} + R_L) [(R_{th} + R_L) - 2R_L]}{(R_{th} + R_L)^4} = 0$$

$$\frac{V_{th}^2 (R_{th} - R_L)}{(R_{th} + R_L)^3} = 0$$

So,

$$R_{th} - R_L = 0$$

$$\boxed{R_{th} = R_L}$$

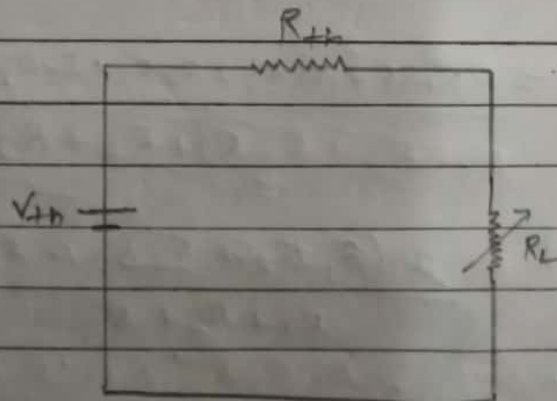
Substituting the value of $R_{th} R_L$ in eqⁿ (ii),
We get —

~~P_{max}~~

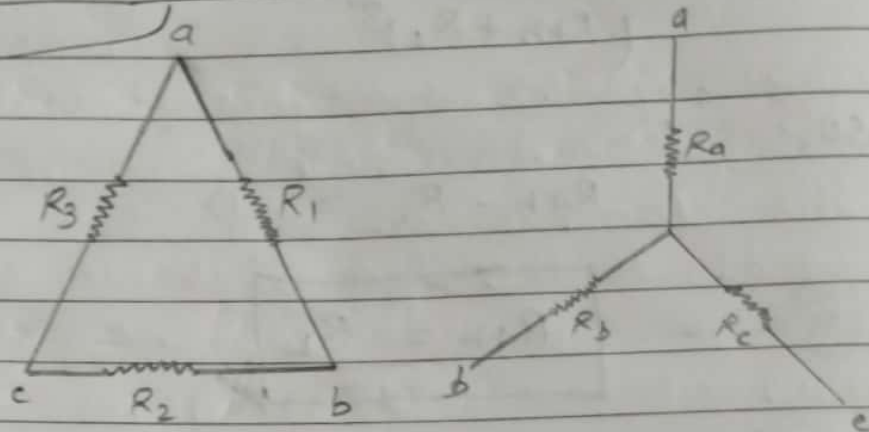
$$P_{max} = \frac{V_{th}^2}{(R_{th} + R_{th})^2} \cdot R_{th}$$

$$P_{max} = \frac{V_{th}^2}{4R_{th}^2} \cdot R_{th}$$

$$\boxed{P_{max} = \frac{V_{th}^2}{4R_{th}}}$$



Delta to star
Star to delta transformation:-



$$R_a + R_b = \frac{R_1(R_2 + R_3)}{R_1 + R_2 + R_3} \quad \text{--- (i)}$$

$$R_b + R_c = \frac{R_2(R_3 + R_1)}{R_1 + R_2 + R_3} \quad \text{--- (ii)}$$

$$R_c + R_a = \frac{R_3(R_2 + R_1)}{R_1 + R_2 + R_3} \quad \text{--- (iii)}$$

Adding eqⁿ (i), (ii) + (iii) —

$$R_a + R_b + R_b + R_c + R_c + R_a = \frac{R_1(R_2 + R_3)}{R_1 + R_2 + R_3} + \frac{R_2(R_3 + R_1)}{R_1 + R_2 + R_3} + \frac{R_3(R_2 + R_1)}{R_1 + R_2 + R_3}$$

$$2(R_a + R_b + R_c) = \frac{R_1R_2 + R_1R_3 + R_2R_3 + R_2R_1 + R_3R_2 + R_3R_1}{R_1 + R_2 + R_3}$$

$$2(R_a + R_b + R_c) = \frac{2(R_1R_2 + R_2R_3 + R_3R_1)}{R_1 + R_2 + R_3}$$

$$R_a + R_b + R_c = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_1 + R_2 + R_3} \quad \text{--- (iv)}$$

Subtracting eqⁿ (i) from eqⁿ (iv).

$$\cancel{R_a} + \cancel{R_b} + R_c - \cancel{R_a} - \cancel{R_b} = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1 + R_2 + R_3} - \frac{R_1(R_2 + R_3)}{R_1 + R_2 + R_3}$$

$$R_c = \frac{\cancel{R_1 R_2} + R_2 R_3 + \cancel{R_3 R_1} - \cancel{R_1 R_2} - \cancel{R_1 R_3}}{R_1 + R_2 + R_3}$$

$$R_c = \frac{R_2 R_3}{R_1 + R_2 + R_3} \quad \text{--- (v)}$$

Similarly - subtract eqⁿ (ii) from eqⁿ (iv)

$$R_a + \cancel{R_b} + \cancel{R_c} - \cancel{R_b} - \cancel{R_c} = \frac{\cancel{R_1 R_2} + \cancel{R_2 R_3} + R_3 R_1 - \cancel{R_2 R_3} - \cancel{R_2 R_1}}{R_1 + R_2 + R_3}$$

$$R_a = \frac{R_3 R_1}{R_1 + R_2 + R_3} \quad \text{--- (vi)}$$

Similarly - subtract eqⁿ (iii) from eqⁿ (iv)

$$\cancel{R_a} + R_b + \cancel{R_c} - \cancel{R_a} - \cancel{R_c} = \frac{R_1 R_2 + \cancel{R_2 R_3} + \cancel{R_3 R_1} - \cancel{R_2 R_2} - \cancel{R_3 R_1}}{R_1 + R_2 + R_3}$$

$$R_b = \frac{R_1 R_2}{R_1 + R_2 + R_3} \quad \text{--- (vii)}$$

Now - start to delta transformation.

multiplying eqn (v) & (vi)

$$R_a \times R_c = \frac{R_2 R_3}{R_1 + R_2 + R_3} \times \frac{R_1 R_3}{R_1 + R_2 + R_3}$$

$$R_a R_c = \frac{(R_1 R_2 R_3)^2}{(R_1 + R_2 + R_3)^2} \quad \text{--- (vii)}$$

multiplying eqn (vi) & (vii)

$$R_a \times R_b = \frac{R_1^2 R_3}{R_1 + R_2 + R_3} \times \frac{R_2 R_1}{R_1 + R_2 + R_3}$$

$$R_a b = \frac{R_1^2 R_2 R_3}{(R_1 + R_2 + R_3)^2} \quad \text{--- (ix)}$$

multiplying eqn (vii) & (v)

$$R_b \times R_c = \frac{R_2 R_1}{R_1 + R_2 + R_3} \times \frac{R_2 R_3}{R_1 + R_2 + R_3}$$

$$R_b R_c = \frac{R_1^2 R_2 R_3}{R_1 + R_2 + R_3} \quad \text{--- (x)}$$

Adding eqn (viii), (ix), & (x) —

$$R_a R_c + R_a R_b + R_b R_c = \frac{R_1 R_2 R_3^2 + R_1^2 R_2 R_3 + R_1 R_2^2 R_3}{(R_1 + R_2 + R_3)^2}$$

$$R_a R_c + R_a R_b + R_b R_c = \frac{R_1 R_2 R_3 (R_3 + R_1 + R_2)}{(R_1 + R_2 + R_3)^2}$$

$$R_a R_c + R_a R_b + R_b R_c = \frac{R_1 R_2 R_3}{R_1 + R_2 + R_3} \quad \text{--- (xi)}$$

Divide eqn (viii) $\div$ v

$$\frac{R_a R_c + R_a R_b + R_b R_c}{R_c} = \frac{R_1 R_2 R_3}{R_1 + R_2 + R_3} \times \frac{R_1 + R_2 + R_3}{R_2 R_3}$$

$$\boxed{R_a + R_b + \frac{R_a R_b}{R_c} = R_1} \quad \text{X}$$

similarly eqn (xi) $\div$ (vi)

$$\frac{R_a R_c + R_a R_b + R_b R_c}{R_a} = \frac{R_1 R_2 R_3}{(R_1 + R_2 + R_3)} \times \frac{(R_1 + R_2 + R_3)}{R_1 R_3}$$

$$\boxed{R_c + R_b + \frac{R_b R_c}{R_a} = R_2} \quad \text{X}$$

similarly: — eqn (xi) $\div$ (vii)

$$\frac{R_a R_c + R_a R_b + R_b R_c}{R_b} = \frac{R_1 R_2 R_3}{(R_1 + R_2 + R_3)} \times \frac{(R_1 + R_2 + R_3)}{R_2 R_1}$$

$$R_a + R_c + \frac{R_a R_c}{R_b} = R_g$$

XX

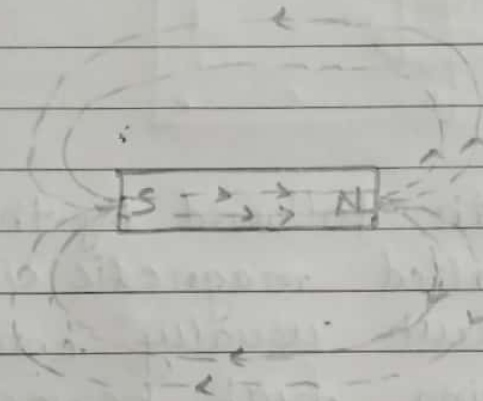
3. Magnetic circuit

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Magnet :- A Material that has a property of Attractive object made up of certain magnetic substances like iron, nickel etc.



Magnetic field :- Space in which magnetic poles experience oppose.



Magnetic field around the magnet bar is represented by imaginary lines called magnetic field lines.

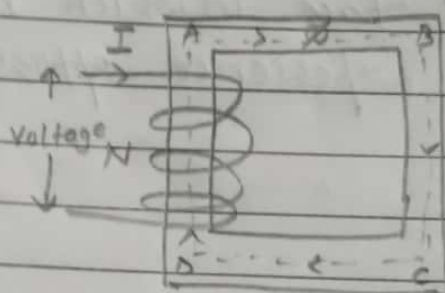
~~It would~~ ^{It would} ~~that~~ emerged from N pole passes through surrounding medium and enters the south pole.

Magnetic flux :- The ~~am~~ Number of magnetic lines produce by the magnetic source. is called ~~mag~~ magnetic flux.

unit - Weber

Permeability :- Its conductivity for magnetic flux.

Magnetic circuits :-



The closed path followed by the magnetic flux is called magnetic circuit. magnetic circuit usually consists of materials having high permeability. Here, magnetic flux followed the path ABCDA and this magnetic circuit.

The amount of flux set up in the coil depends upon current and No. of turns

$$NI = \text{MMF}$$

(magnetomotive force)

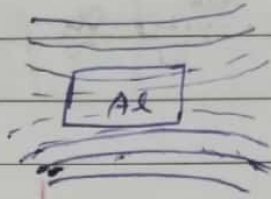
Magnetic material :-

- (i) Paramagnetic material ^{slightly} - ex, Al, sodium, platinum etc.
- (ii) Diamagnetic material ^{repelled} - ex - Cu, gold, silver etc.
- (iii) Ferromagnetic material ^{highly attractive} - ex - Iron

Types of magnetic material -

(1) Paramagnetic material:- which are weakly attractive towards the magnetic field. ex- Aluminium, sodium, platinum etc.

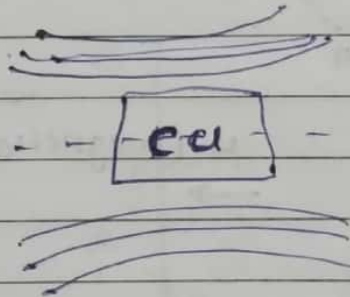
(ii) ~~di~~



Permeability $\rightarrow$ slightly more than 1.

(2) Diamagnetic material:-

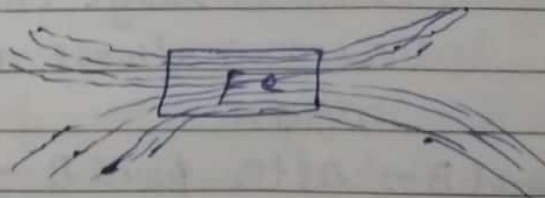
which are repelled by the magnetic field. ex- copper, gold, silver ~~etc~~, hydrogen etc.



Permeability is $0 < \mu < 1$.

3. Ferromagnetic material:-

which are strongly attractive ~~towards~~ ^{by} the magnetic field. ex- Iron, Cobalt, Nickel etc.



Permeability is $100 < \mu < 3500$.

Relative permeability - It is the ratio of

$$\mu = \mu_r \mu_0$$

$$\mu_r = \frac{\mu}{\mu_0}$$

Note:-

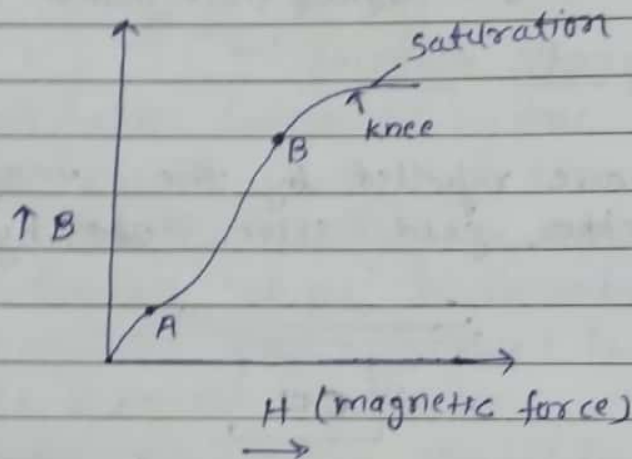
ϕ = Magnetic flux

B = Magnetic flux density

H → Magnetising force

#

B-H curve:-



$$H = \frac{NI}{l}$$

It is a graphical representation of relation b/w B (magnetic flux) and applied magnetising force.

OA Region - B increases gradually as the value of H increases.

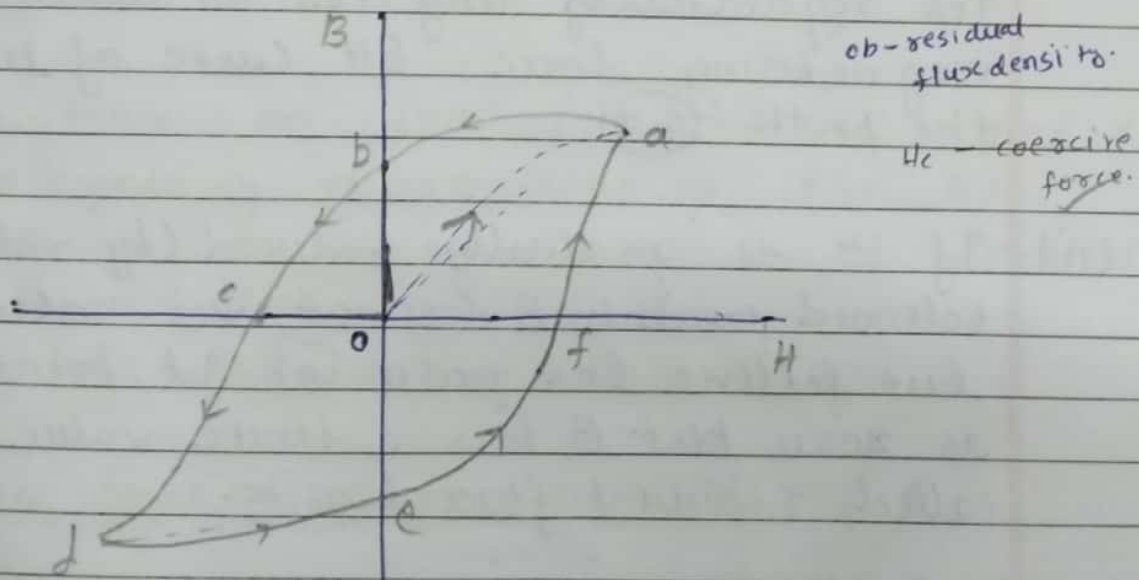
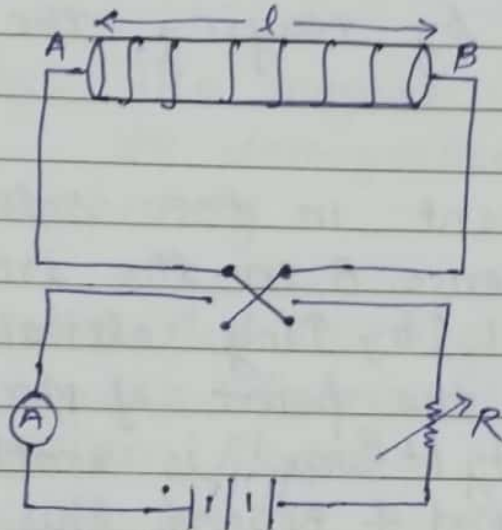
AB Region - For a small change in H there is large change in B . B-H curve is almost linear in this region.

Beyond B - After point B the change in magnetic flux density (B) is small even for a large change in H .

Finally

In H. The B-H curve will be parallel to x-axis. This region is saturation region.

Magnetic Hysteresis -



The phenomenon of lagging of magnetic flux density (B) behind the magnetising force (H) in a magnetic material, subjected to a cycle of ~~mag~~ magnetisation. (When magnetic material is subjected to a cycle of magnetisation)

i.e. it is magnetised first in one direction & then in the other).

consider an unmagnetised iron bar AB wound with N turn. The magnetising force

$H = NI/l$ produced by this solenoid can be changed by varying the current through the coil—

(i) When current in the solenoid is zero, $H=0$ & hence B in the iron is zero. As H is $\uparrow$ ed (by $\uparrow$ ing solenoid current), B also $\uparrow$ es until the point of max. magnetic flux density ($+B_{max}$) is reached. The material is saturated & beyond this point B will not $\uparrow$ es regardless of any $\uparrow$ es in current or magnetising force. BH curve of iron follows the path (oa) .

(iv)

(ii) If H is gradually reduced (by reducing solenoid current), B does not $\downarrow$ es along oa but follows the path (ab) . At point (b) H is zero but B has a finite value (ob) called residual flux density.

☆

Retentivity:— The property of a magnetic material to retain its magnetic character even after removal of magnetising field.

#

(iii) To remove residual magnetism ob , H is reversed by reversing the current through

the coil. When H is fed in the reverse direction, the $B-H$ curve follows the path (bc) so that when $H = 0$, the residual magnetism is zero. The value of $H = 0$ required to wipe out residual magnetism called H_c (coercive force).

coercivity: The magnetising field needed to demagnetise the magnetic material completely.

- (iv) If H is further fed in negative direction material again saturated at point (d) in -ve direction. Reducing H to zero & then feeding it in the +ve direction completes the curve defn.

$\therefore$ When an iron piece is subjected to one cycle of magnetisation, the BH curve is a closed loop called hysteresis loop.

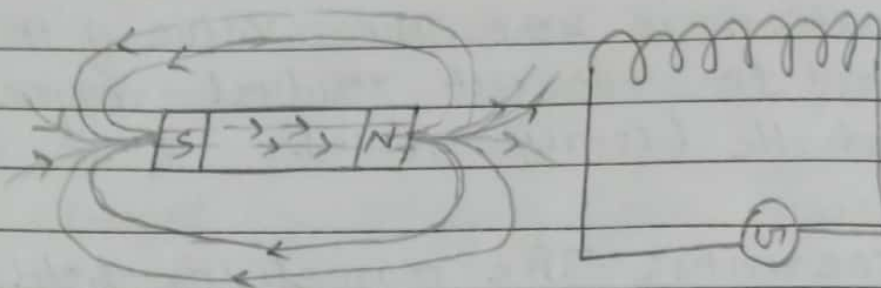
* Basic laws of electromagnetism:-

Faraday's first law of Electromagnetic induction:-

First law:- Whenever the amount of magnetic flux linked with a circuit changes, an e.m.f is induced in the circuit. The induced e.m.f exists so long

as the change in magnetic flux continues.

~~Second law~~



When bar magnet move towards away coil there will be change in flux, emf will be induced & galvanometer shows deflection.

Second law :-

The magnitude of emf induced in a circuit is directly proportional to the rate of change of magnetic flux linked with the circuit.

$$\text{induced emf } e \propto \frac{(\phi_2 - \phi_1)}{\Delta t}$$

$$e = K \frac{(\phi_2 - \phi_1)}{\Delta t}$$

Here $K=1$ in all system of units.

$$e = \frac{\phi_2 - \phi_1}{\Delta t}$$

If $d\phi$ is the small change in magnetic

flux in a small time dt then we can write —

$$e = - \frac{d\phi}{dt}$$

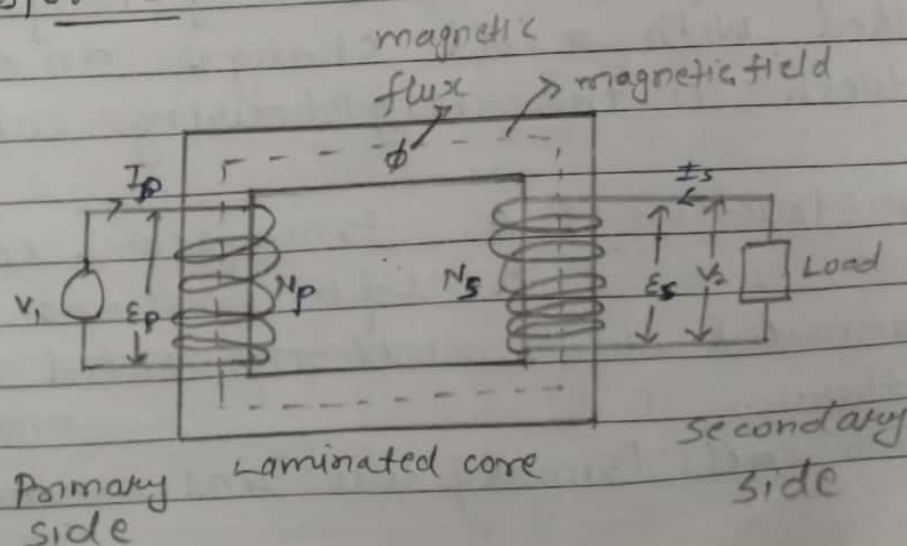
* Negative sign is taken because induced e.m.f always opposes any change in magnetic flux associated with the circuit.

In case of a closely wound of N turns, change in magnetic flux associated with each turn is the same. Therefore total induced e.m.f is given by —

$$e = -N \frac{d\phi}{dt}$$

So, by increasing the number of turns N in the coil we can increase the induced e.m.f.

Transformer :- (1- ϕ)



Transformer:- Transmore is a static device which transfer the electrical energy one circuit to another circuit without changing the frequency.

or based on voltage types of transformer

- (i) step up
- (ii) step down

Transformer:- A transformer is an electrical device which is used for changing the A.C voltages.

A transformer which increases the A.C voltages is called step up transformer and a transformer which decreases the A.C voltages is called a step down transformer.

Principle:- A transformer is based on the principle of mutual induction. i.e whenever the amount of magnetic flux linked with a coil changes, an e.m.f is induced in the neighbouring coil.

Construction - A transformer consist of a soft iron core made of laminated sheets, well insulated from each other.

Two coils Primary coil and secondary coil

are wound on the same core, but are well insulated from each other. Note that both the coils are also insulated from the core.

The source of alternating e.m.f (to be transformed) is connected to the secondary coil.

Theory & working :-

Let the alternating e.m.f supplied by the

$$E_p = E_0 \sin \omega t$$

The alternating primary current induces an alternating magnetic flux Φ_B in the iron core. Because the core extends through the turns of secondary, so the induced flux also extends through the secondary winding.

According to the faraday's law of electromagnetic induction the induced e.m.f per turn (E_{turn}) is same for both primary & secondary winding.

$$E_{\text{turn}} = \frac{d\Phi_B}{dt} = \frac{E_p}{N_p} = \frac{E_s}{N_s}$$

Here, N_p, N_s represents the total number of turns in the primary and secondary coils respectively -

$$\frac{E_s}{E_p} = \frac{N_s}{N_p}$$

As we assume that no energy is lost along the way, conservation of energy requires that-

$$\boxed{\text{Input Power} = \text{output Power}}$$

$$I_p E_p = I_s E_s \quad \therefore \frac{I_p}{I_s} = \frac{E_s}{E_p}$$

In general,-

$$\boxed{\frac{E_s}{E_p} = \frac{I_p}{I_s} = \frac{N_s}{N_p} = K}$$

Here, $K = \text{transformation ratio}$

Note:-

If $N_s > N_p$ then $E_s > E_p$ then the transformer is called step up transformer.

If $N_s < N_p$ then $E_s < E_p$ then the transformer is called a step down transformer.

Efficiency of transformer:-

It is defined as the ratio of output power to the input power.

$$\boxed{\eta = \frac{\text{Output power}}{\text{Input power}} \times 100\% = \frac{E_s I_s}{E_p I_p} \times 100\%}$$

NOTE:- The transformer cannot work on D.C. A transformer changes A.C voltages/currents. It does not affect the frequency of A.C. When A.C voltage is raised n times the corresponding alternating current reduces to $1/n$ times.

Energy Losses in a transformer :-

- (i) Copper Loss :- It is the energy loss in the form of heat in the copper coils of a transformer. This is due to joule heating of conducting wires. These are minimised using thick wires.
- (ii) Iron Loss :- It is the energy loss in the form of heat in the iron core of the transformer. This is due to formation of eddy currents in iron core. It is minimised by taking laminated cores.
- (iii) Leakage of magnetic flux :-

Due to this the rate of change of magnetic flux linked with each turn of S_1, S_2 is less than the rate of change of magnetic flux linked with each turn of P_1, P_2 . It can be reduced by winding the primary and secondary coils one over the other.

(iv) Hysteresis Loss :- This is the loss of energy due to repeated magnetisation and demagnetisation of the iron core when A.C. is fed to it. The loss is kept to a minimum by using a magnetic material which has a low hysteresis loss.

(v) Magnetostriction :- It is humming noise of a transformer. Therefore output power in the best transformer may be roughly 90% of the input power.

EMF equation of the transformer :-

Let,

ϕ_m be the max^m value of flux in weber

f be the supply frequency in Hz

N_1 is the number of turns in the primary winding

N_2 is the number of turns in the secondary winding —

By faradays law —

Let E_1 be the emf induced in the primary winding —

$$E_1 = - \frac{d\psi}{dt} \quad \dots \dots (1)$$

Where, $\psi = N_1 \phi$

$$E_1 = - N_1 \frac{d\phi}{dt}$$

Since ϕ is due to AC supply $\phi = \phi_m \sin \omega t$

$$E_1 = - N_1 \frac{d}{dt} (\phi_m \sin \omega t)$$

$$E_1 = - N_1 \omega \phi_m \cos \omega t$$

$$E_1 = N_1 \omega \phi_m \sin(\omega t - \pi/2)$$

So the induced emf lags flux by 90° .

Maximum value of emf

$$E_{1, \max} = N_1 \omega \phi_m$$

We know that

$$\omega = 2\pi f$$

$$E_{1, \max} = 2\pi f N_1 \phi_m$$

Root mean square RMS value is —

$$E_1 = \frac{E_{1, \max}}{\sqrt{2}} \quad \text{--- (2)}$$

putting the value of $E_{1, \max}$ in equation (2) —

$$E_1 = \sqrt{2} \pi f N_1 \phi_m$$

$$E_1 = 1.414 \times 3.14 \times f N_1 \phi_m$$

$$E_1 = 4.44 f N_1 \phi_m \quad \text{--- (3)}$$

Similarly —

$$E_2 = \sqrt{2} \pi f N_2 \phi_m$$

$$E_2 = 4.44 f N_2 \phi_m \quad \text{--- (4)}$$

from eqn (3) & (4)

$$\frac{E_2}{E_1} = \frac{4.44 f N_2 \phi_m}{4.44 f N_1 \phi_m}$$

$$\frac{E_2}{E_1} = \frac{N_2}{N_1} = k$$

The above equation is called the turn ratio where k is known as the transformation ratio.

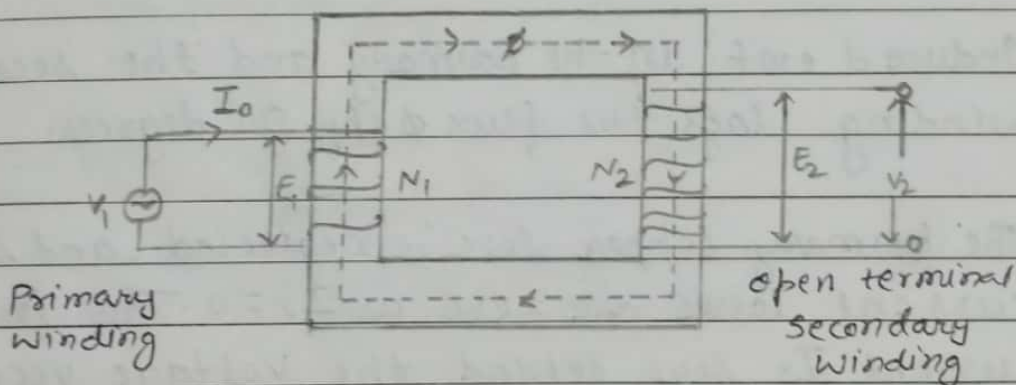
For sinusoidal wave —

$$\frac{\text{R.M.S value}}{\text{Average value}} = \text{Form factor} = 1.11$$

Here 1.11 is the form factor.

Transformer on load and no load:-

* Transformer on No load:-



When the primary of a transformer is connected to the source of ac supply and the secondary is open, the transformer is said to be at no load. (there is no load on secondary) while primary winding carries a small current I_0 called no-load current which is 2 to 10% of the rated current.

* The no load current consists of two components:-

• Reactive or magnetizing component I_m :-

It is ~~an~~ in quadrature with the applied voltage V_1 . it produces flux in the core and does not consume any power.

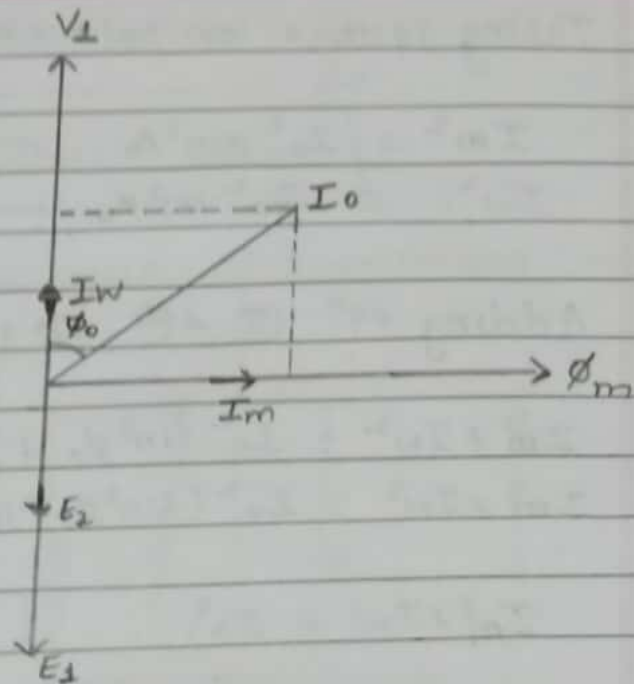
- Active or Power component I_w

Also known as a working component :

(It is in phase with the applied voltage V_1 . It supplies the iron losses and a small amount of primary copper loss.)

* The following steps are given below to draw the phasor diagram:—

- (i) The function of the magnetizing component is to produce the magnetizing flux, and thus, it will be in phase with the flux.
- (ii) Induced emf in the primary and the secondary winding lags the flux ϕ by 90 degrees.
- (iii) The primary copper loss is neglected, and secondary current losses are zero as $I_2 = 0$. Therefore, the current I_0 lags behind the voltage vector V_1 by an angle ϕ_0 called the no-load power factor angle and is shown in the phasor diagram.
- (iv) The applied voltage V_1 is drawn equal and opposite to the induced emf E_1 , because the difference b/w the two, at no load, is negligible.
- (v) Active component I_w is drawn in phase with the applied voltage V_1 .
- (vi) The phasor sum of magnetizing current I_m and the working current I_w gives the no-load current I_0 .



we know that

$$\frac{E_1}{E_2} = \frac{N_1}{N_2} = K$$

$$\therefore E_2 = E_1 \times \frac{N_2}{N_1}$$

$$\boxed{\therefore E_2 = \frac{E_1}{\alpha}}$$

From phase diagram, we can derive —

Let,

ϕ_0 = no load power factor angle

$\therefore \cos \phi_0$ = no load power factor.

$$I_m = I_0 \sin \phi_0 \quad \text{--- (i)}$$

$$I_w = I_0 \cos \phi_0 \quad \text{--- (ii)}$$

Taking square on both sides of eqⁿ (i) & (ii)

$$I_m^2 = I_o^2 \sin^2 \phi_o \quad \text{--- (3)}$$

$$I_w^2 = I_o^2 \cos^2 \phi_o \quad \text{--- (4)}$$

Adding eqⁿ (3) & (4), we get,

$$I_m^2 + I_w^2 = I_o^2 \sin^2 \phi_o + I_o^2 \cos^2 \phi_o$$

$$I_m^2 + I_w^2 = I_o^2 (\sin^2 \phi_o + \cos^2 \phi_o)$$

$$I_m^2 + I_w^2 = I_o^2 \quad \left\{ \sin^2 \phi_o + \cos^2 \phi_o = 1 \right\}$$

$$I_o = \sqrt{I_m^2 + I_w^2} \quad \text{--- (5)}$$

from eqⁿ (2)

$$\cos \phi_o = \frac{I_w}{I_o} \quad \text{--- (6)}$$

$$\text{Core loss} = V_1 I_o \cos \phi_o$$

$$\text{But } \cos \phi_o = \frac{I_w}{I_o}$$

$$\therefore \text{Core loss} = V_1 \cancel{I_o} \times \frac{I_w}{\cancel{I_o}}$$

$$\text{Core loss} = V_1 I_w$$

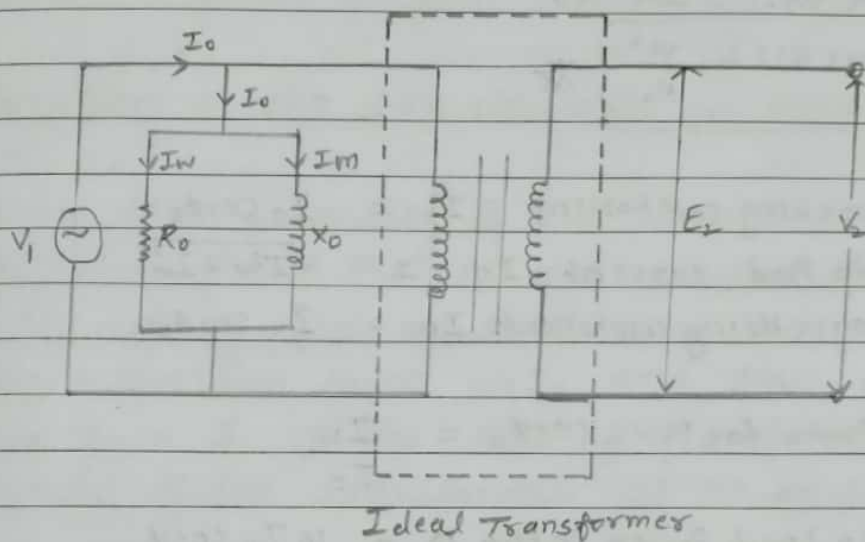
$$\text{Magnetizing Volt ampere} = V I_o \sin \phi_o$$

$$\text{But } \sin \phi_o = \frac{I_m}{I_o}$$

$$\therefore \text{magnetizing voltampere} = V_1 I_0 \times \frac{I_u}{I_0}$$

$$\text{magnetizing voltampere} = V_1 I_u$$

* No Load equivalent circuit of transformer :-



$$V_1 = I_w R_0 \quad \text{--- (1)}$$

$$V_1 = I_m X_0 \quad \text{--- (2)}$$

$$\therefore V_1^2 = I_w^2 R_0^2 \quad \text{--- (3)}$$

$$V_1^2 = I_m^2 X_0^2 \quad \text{--- (4)}$$

$$\therefore I_w^2 = \frac{V_1^2}{R_0^2} \quad \text{--- (5)}$$

$$I_m^2 = \frac{V_1^2}{X_0^2} \quad \text{--- (6)}$$

Adding eqn (5) + (6)

$$I_w^2 + I_m^2 = \frac{V_1^2}{R_0^2} + \frac{V_1^2}{X_0^2}$$

$$I_w^2 + I_m^2 = I_0^2$$

$$\therefore I_0^2 = I_w^2 + I_m^2$$

$$I_0 = \sqrt{I_w^2 + I_m^2} \quad \dots (7)$$

$$\text{Core loss} = I_w^2 R_0$$

$$\text{Core loss} = \frac{V_1^2}{R_0} \quad \times$$

$$\text{Working component} = I_w = I_0 \cos \phi_0$$

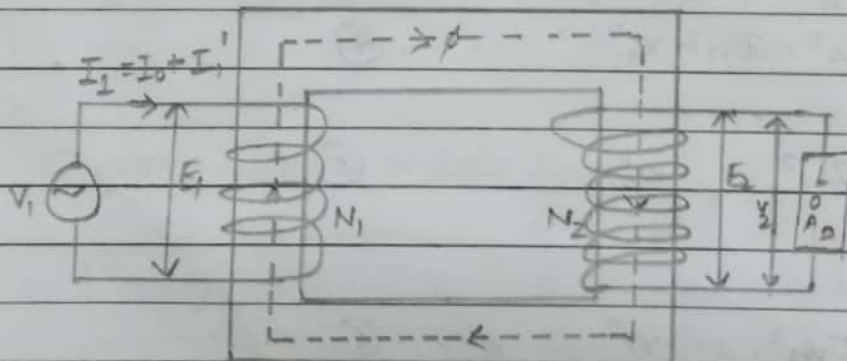
$$\text{No load current } I_0 = \sqrt{I_w^2 + I_m^2}$$

$$\text{Magnetizing component } I_m = I_0 \sin \phi_0$$

$$\text{Power factor } \cos \phi_0 = \frac{I_w}{I_0}$$

$$\text{No load Power input } P_0 = V_1 I_0 \cos \phi_0$$

Transformer on load condition



When the transformer is on the loaded condition, the secondary of the transformer is connected to load. The load can be resistive

inductive or capacitive. The current I_2 flows through the secondary winding of the transformer. The magnitude of the secondary current depends on the terminal voltage V_2 and the load impedance. The phase angle b/w the secondary current and voltage depends on the nature of the load.

- Operation of the transformer on load condition:-
- When the secondary of the transformer is kept open, it draws the no-load current from the main supply. The no load current induces the magnetomotive force $N_0 I_0$ and this force set up the flux ϕ in the core of transformer. The circuit of the transformer at no load condition.
- When the load is connected to the secondary of the transformer, I_2 current flows through their secondary winding. The secondary current induces the magnetomotive force $N_2 I_2$ on the secondary winding of the transformer. This force set up the flux ϕ_2 in the transformer core. The flux ϕ_2 opposes the flux ϕ according to lenz's law.
- As the flux ϕ_2 opposes the flux ϕ , the resultant flux of the transformer decreases and this flux reduces the induced EMF E_1 . Thus, the strength of the V_1 is more than E_1 and an additional primary current I_1' drawn from the main supply. The additional current is used for restoring

the original value of the flux in the core of the transformer so that $V_1 = E_1$.

The primary current I_1' is in phase opposition with the secondary current I_2 . Thus it is called the primary counter-balancing current.

- The additional current I_1' induces the magnetomotive force $N_1 I_1'$. And this force set up the flux ϕ_1 . The direction of the flux is the same as that of the ϕ and it cancels the flux ϕ_2 which induces because of the MMF $N_2 I_2$.

Now:-

$$N_1 I_1' = N_2 I_2$$

$$\therefore I_1' = \left(\frac{N_2}{N_1} \right) I_2 = K I_2$$

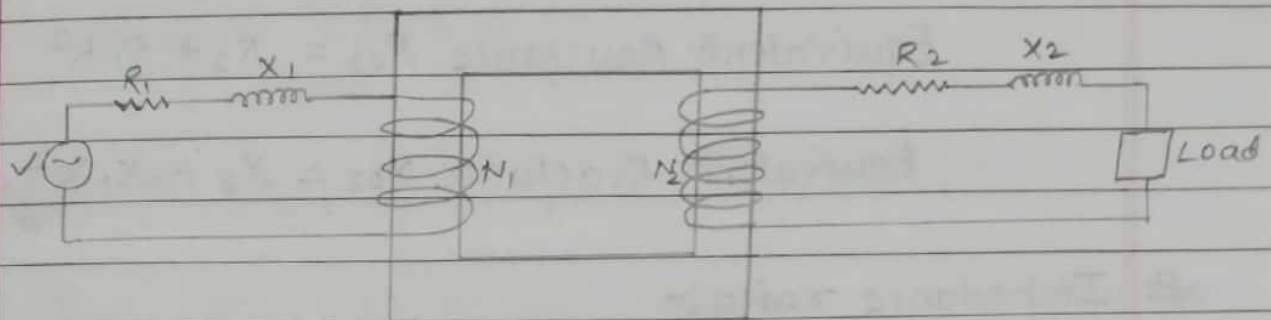
- The phase difference b/w V_1 and I_1 gives the power factor angle ϕ_1 of the primary side of the transformer.
- The power factor of the Secondary side depends upon the type of load connected to the transformer.
- If the load is inductive as shown in the above phasor diagram, the power factor will be lagging and if the load is capacitive the power factor will be leading. The total primary current I_1 is the vector sum of the currents I_0 and I_1' .

i.e. -

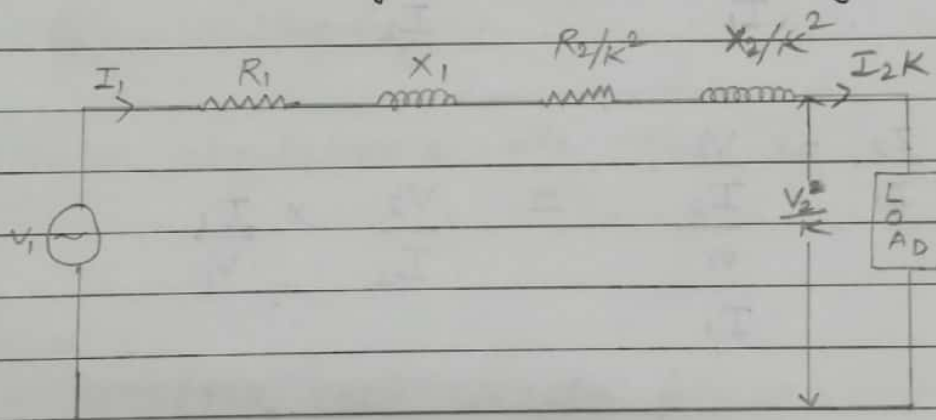
$$I_1 = I_0 + I_1'$$

Equivalent circuit of transformer

Representation of any device with the help of active (P, I, V) & passive element (R, L, C) that can be used for analysis and predict the performance of the device.



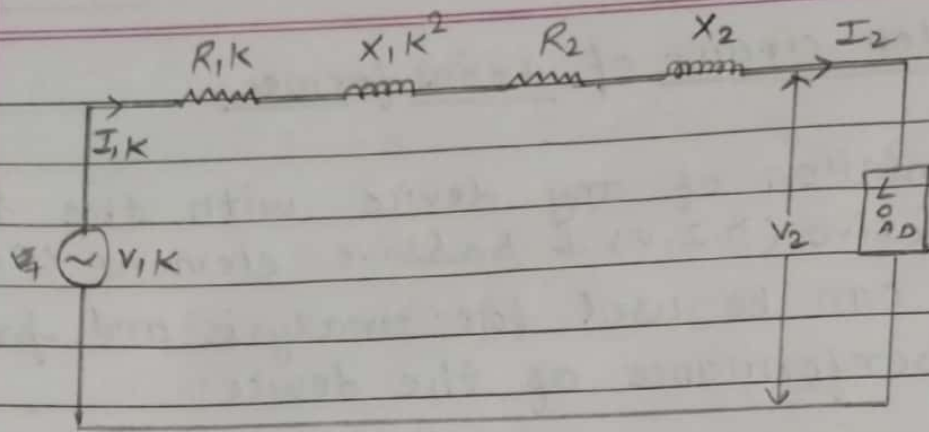
(i) Equivalent ckt referred to primary side—



Equivalent Resistance = $R_{01} = R_1 + R_2/K^2$

Equivalent Reactance = $X_{01} = X_1 + X_2/K^2$

(ii) Equivalent ckt referred to secondary side—



Equivalent Resistance $R_{02} = R_2 + R_1 k^2$

Equivalent Reactance $X_{02} = X_2 + X_1 k^2$

Impedance ratio:-

$$Z_1 = \frac{V_1}{I_1}, \quad Z_2 = \frac{V_2}{I_2}$$

$$\frac{Z_2}{Z_1} = \frac{\frac{V_2}{I_2}}{\frac{V_1}{I_1}} = \frac{V_2}{I_2} \times \frac{I_1}{V_1}$$

$$\frac{Z_2}{Z_1} = \frac{V_2}{V_1} \times \frac{I_1}{I_2} \quad \left\{ \because \frac{V_2}{V_1} = \frac{I_1}{I_2} = k \right\}$$

$$\frac{Z_2}{Z_1} = k \times k$$

$$\boxed{\frac{Z_2}{Z_1} = k^2}$$

as we know:-

$$Z = \sqrt{R^2 + X^2}$$

impedance consist of resistance & Reactance
Similarly;

$$\frac{X_2}{X_1} = K^2, \quad \frac{R_2}{R_1} = K^2$$

voltage regulation of transformer:-

It is the arithmetic difference b/w the No load secondary voltage and secondary voltage on load express as a percentage of No load Secondary voltage.

$$\text{Voltage regulation} = \frac{V_{20} - V_2}{V_{20}} \times 100$$

Numerical:-

Q 10 KVA 2000/400 volt single phase transformer has:- $R_1 = 5 \Omega$, $R_2 = 0.2 \Omega$, $X_1 = 12 \Omega$, $X_2 = 0.48 \Omega$ and $\cos \phi = 0.8$ lagging. determine the equivalent resistance and reactance referred to secondary side. Also determine voltage drop.

Solⁿ:-

$$\frac{R_2}{R_1} = K^2 \Rightarrow \frac{0.2}{5025} = 0.04$$

$$R_{02} = R_2 + R_1 K^2$$

$$R_{02} = 0.2 + \frac{8 \times 0.04}{100255}$$

$$R_{02} = 0.2 + 0.2$$

$$R_{02} = 0.4 \Omega$$

$$X_{02} = X_2 + X_1 K^2$$

$$= 0.48 + \frac{12 \times 0.04}{10025}$$

$$= 0.48 + 0.48$$

$$X_{02} = 0.96$$

Voltage drop —

$$V_{02} = I_2 R_{02} \cos \phi_2 - I_2 X_{02} \sin \phi_2$$

$$\cos \phi_2 = 0.8$$

$$\sin \phi_2 = \sqrt{1 - 0.64}$$

$$\sin \phi_2 = \sqrt{0.36}$$

$$\sin \phi_2 = 0.6$$

$$\text{Voltage drop} = 25 (0.4 \times 0.8 + 0.96 \times 0.6)$$

$$= 22.4 \text{ volt Ans.}$$

Losses in transformer

Losses appear in the form of heat and produce first increase in temperature, (2) drop in efficiency

There are two types of losses:—

- (i) Iron loss / core loss
- (ii) Copper loss

(i) Iron loss :- It occurs in the transformer core due to Alternating flux it consists of hysteresis and eddy current loss.

(a) Hysteresis loss :- loss of energy due to continuous magnetization and de-magnetization of transformer.

$$\text{Hysteresis loss} = K_h B^{1.6} f \quad \text{Watt/m}^3$$

(b) Eddy current loss :- When a conductor is placed in a moving magnetic field loops of electric current are induced in it. These circulating current further produces eddy current loss.

$$\text{eddy loss} = K_e B^2 f^2 T^2$$

where T = thickness of conductor

unit = Watt/m³

NOTE :- iron losses determine by open circuit test.

(2) Copper loss :- This loss occurs in both primary winding and secondary winding due to their ohmic resistance.

Note :- Copper losses determine by short circuit test.

*

$$P_i = \text{Hysteresis loss} + \text{eddy loss}$$

*

$$P_c = I_1^2 R_1 + I_2^2 R_2$$

Efficiency of a transformer :-

Efficiency is the ratio of output power to the input power.

condition for maximum efficiency :-

$$\text{efficiency} = \frac{\text{o/p power}}{\text{i/p power}}$$

$$\eta = \frac{\text{o/p power}}{\text{o/p power} + \text{Total losses}}$$

$$\eta = \frac{V_2 I_2 \cos \phi_2}{V_2 I_2 \cos \phi_2 + P_i + P_c}$$

$$\eta = \frac{V_2 I_2 \cos \phi_2}{V_2 I_2 \cos \phi_2 + P_i + I_2^2 R_{02}}$$

$$\eta = \frac{I_2 (V_2 \cos \phi_2)}{I_2 (V_2 \cos \phi_2 + \frac{P_i}{I_2} + I_2 R_{02})}$$

To get max efficiency -

Denominator should be minimum -

$$\frac{d}{dI_2} (\text{denominator}) = 0$$

$$\frac{d}{dI_2} \left(V_2 \cos \phi_2 + \frac{P_i}{I_2} + I_2 R_{02} \right) = 0$$

$$0 + \frac{d}{dI_2} \left(\frac{P_i}{I_2} \right) + \frac{d}{dI_2} (I_2 R_{02}) = 0$$

$$-\frac{P_i}{I_2^2} + R_{02} = 0$$

$$-P_i + I_2^2 R_{02} = 0$$

$$P_i = I_2^2 R_{02}$$

$$\boxed{P_i = P_c} \quad \text{**}$$

When iron loss is equal to copper loss then the max efficiency observed.

Numerical:-

Q. 440/110 volt transformer has a R_1 is 0.03Ω & R_2 is 0.02Ω iron loss $P_i = (150 \text{ W})$, Determine the secondary current at which unity maximum efficiency will occur at unity power factor ($\cos \phi = 1$). also calculate the value of this max^m efficiency at unity power factor load.

Solⁿ:-

Given:-

$$440/110 \text{ V}$$

$$R_1 = 0.03 \Omega, R_2 = 0.02 \Omega$$

$$P_i = 150 \text{ W}$$

$$\frac{R_2}{R_1} = K^2$$

$$K^2 = \frac{0.02}{0.03}$$

$$K^2 = 0.66$$

$$R_{02} = R_2 + R_1 K^2$$

$$R_{02} = 0.02 + 0.03 \times 0.66$$

$$R_{02} = 0.02 + 0.0198$$

$$R_{02} = 0.0398$$

$$P_i = P_c$$

$$150 = I_2^2 \times 0.0398$$

$$I_2^2 = \frac{150}{0.0398}$$

$$I_2^2 = 3768.844$$

$$I_2 = \sqrt{3768.844}$$

$$I_2 = 61.39 \text{ Amp.}$$

$$\text{Efficiency} = \frac{\text{o/p power}}{\text{o/p power} + \text{total loss}}$$

$$= \frac{V_2 I_2 \cos \phi_2}{V_2 I_2 \cos \phi_2 + P_i P_c}$$

$$= \frac{110 \times 61.39 \times 1}{110 \times 61.39 + 300}$$

$$= \frac{6753}{7053} = 0.95 \times 100$$

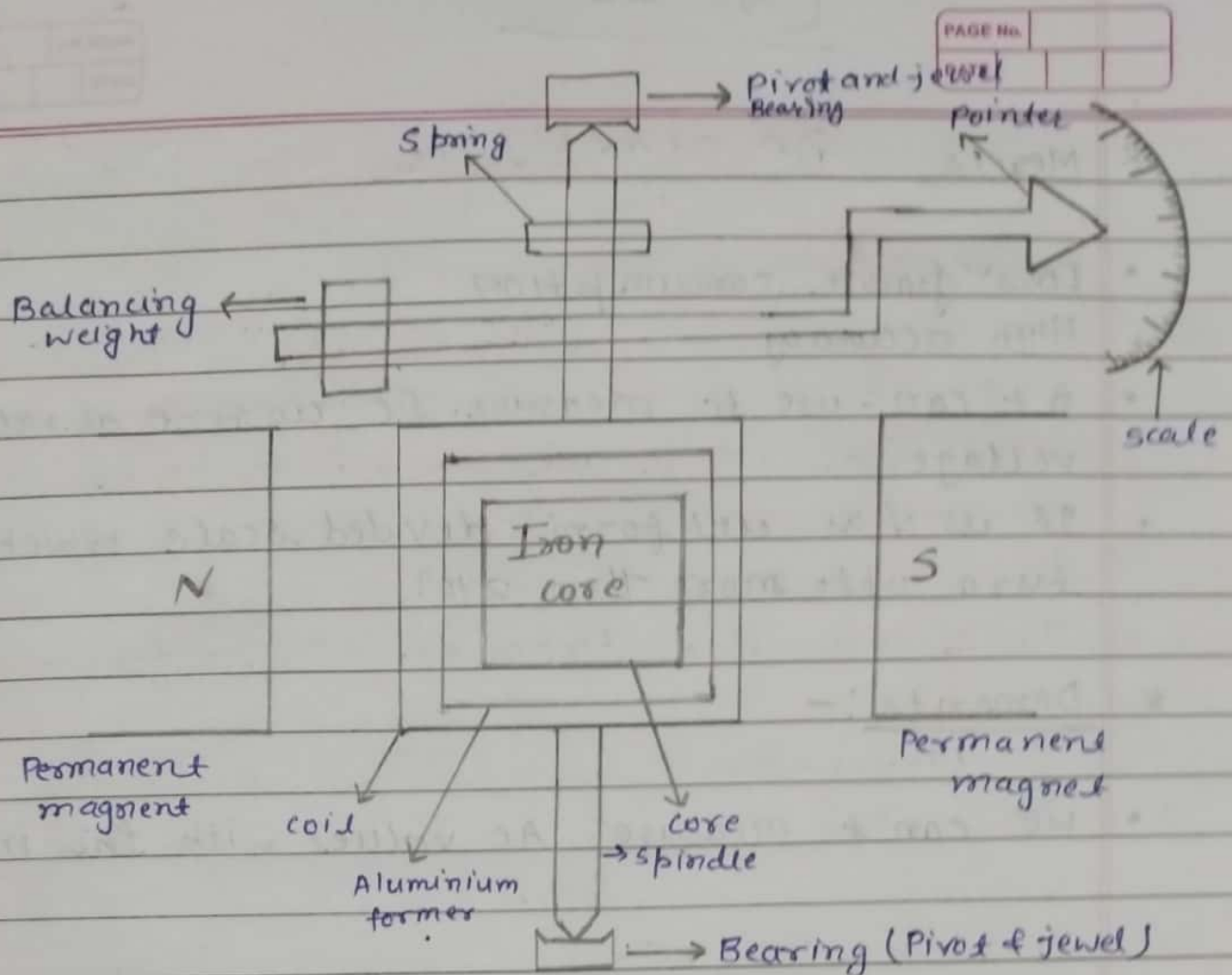
$$= 95\%$$

(Instrument)

PMMC :- Permanent Magnet Moving coil

These are use either as Ammeter or voltmeter and a suitable for DC work (DC voltage & DC current) only.

When a current carrying conductor is placed in a magnetic field then mechanical force. Hence torque acts on the conductor.



Construction:- It consists of rectangular coil of many turns of fine wire wound on an aluminium former inside which is an iron core. This core is placed between the poles of a permanent magnet and the pointer is further attached to the iron coil.

Working:- When the instrument is connected to measure current, the operating current flows to the coil and is placed in the magnetic field of the permanent magnet. The force of mechanical force hence torque acts on it as a result the pointer attached to the moving system moves in a clockwise direction to indicate the value of current.

* Merits

- Less power consumption
- High accuracy
- It can use to measure DC current as well as voltage.
- It has uniform divided scale which can turn upto more than 270° .

* Demerits :-

- We can't measure AC values with this instrument.

DC MACHINEDC Machine :-

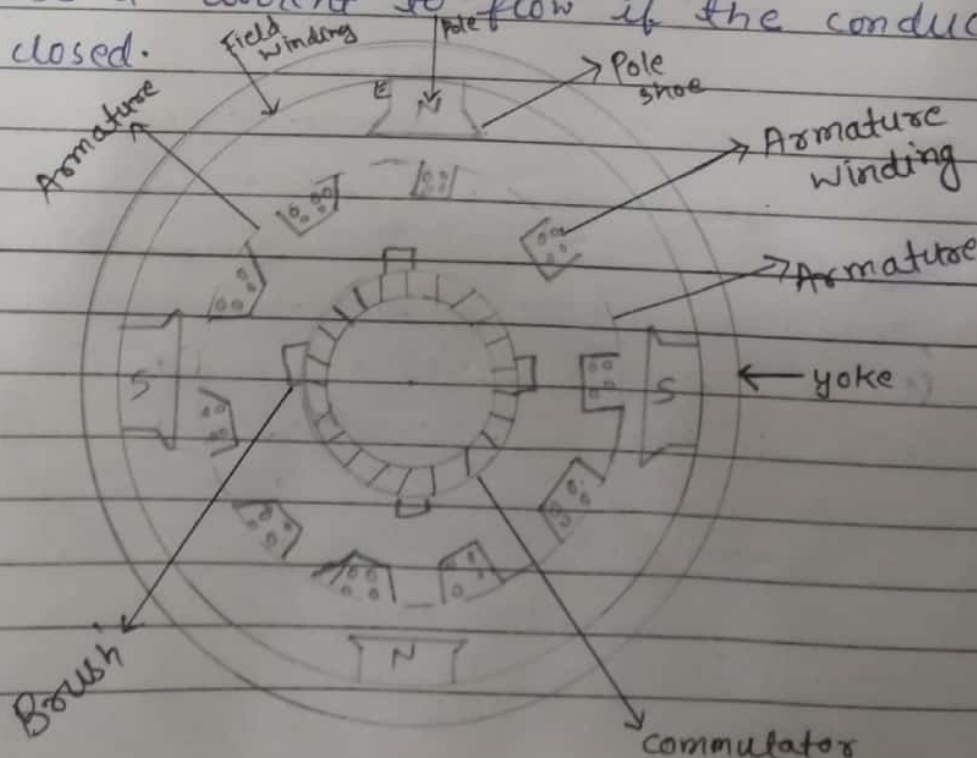
A device which converts electric power into mechanical power or mechanical power into electric power is called DC machine.

* There are two types of DC machine

- (i) DC generator
- (ii) DC motor

(i) DC generator :-

Generator is a device / Machine that converts Mechanical energy into electrical energy and it works in the principle of when a flux cut by the conductor and emf is induced which will cause a current to flow if the conductor circuit is closed.



Field system: - The function of field system is to provide uniform magnetic field within which armature rotates. It consist of No. of poles bolted inside circular frame (yoke). The yoke is usually ^{made of} cast steel. Field coils are mounted on the poles and carried DC current. The MMF when DC current passes through field coil. MMF develop by the field coil produce magnetic flux that passes through the pole pieces the air gap and the armature.

Whenever these flux spread out that is considered magnetic field and we know when a current carrying conductor is placed in a magnetic field force is produced. Hence torque develop due to this torque Armature starts rotating and when these flux cut by the armature conductor which is placed in the Armature slot emf is induced in the Armature part.

The area b/w the poles and armature is ranging from 0.5 to 1.5 mm.

When emf is develop by Armature with the help of commutators. AC voltage is converted into DC voltage and with the help of brushes DC voltage will be collected and supply to the external load.

Armature:- Armature core & Armature winding
This Armature core is made from high permeability material that is silicon steel and this silicon steel is made of No. stamping. And each stamping is separated from its neighbouring one by thin putting of varnish as insulation.

Armature winding:- The slots of armature core hold insulated conductors known as armature winding and this is the winding where working emf is induced.

Emf equation of generator :-

In one revolution, armature cuts total flux produced by all poles -

$$\text{change in flux } (d\phi) = P\phi$$

Time taken to complete one revolution.

$$dt = \frac{60}{N}$$

$$\text{emf of conductor} = \frac{d\phi}{dt} = \frac{P\phi}{\frac{60}{N}} = \frac{P\phi N}{60}$$

$$E_g = \text{Emf of generator (emf per parallel path)}$$

$$= \frac{\text{emf}}{\text{No. of conductors}} \times \frac{\text{No. of conductors}}{\text{No. of parallel path}}$$

$$E_g = \frac{P\phi N}{60} \times \frac{Z}{A}$$

$$E_g = \frac{P\phi Z N}{60A}$$

{ 10

Q. 6 pole left wound DC generator has 600 conductor on its armature the flux per pole is 0.02 weber. calculate the speed at which the generator must be run to generate 300 volt.

- at lap winding
- wave winding

Given:-

$$E = 300 \text{ V}$$

$$\phi = 0.02 \text{ Wb}$$

$$P = 6$$

$$Z = 600$$

In case of lap $\Rightarrow (A = P = 6)$

$$E_g = \frac{P \phi Z N}{60 A}$$

$$300 = \frac{6 \times 0.02 \times 600 \times N}{60 \times 6}$$

$$N = \frac{300 \times 60 \times 6 \times 100}{6 \times 0.02 \times 600}$$

$$N = 1500 \text{ rpm}$$

radian per minute.

In case of wave winding — $(A = 2)$

$$E_g = \frac{P \phi Z N}{60 A}$$

$$N = \frac{60 \times E_g A}{P \phi Z}$$

$$N = \frac{60 \times 300 \times 2 \times 100}{8 \times 0.02 \times 600}$$

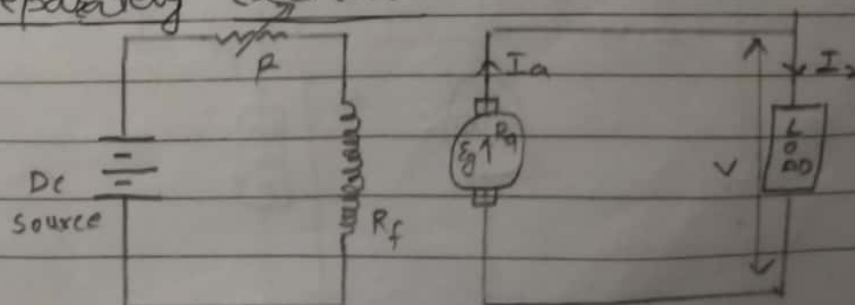
$$N = 500 \text{ rpm}$$

Types of DC Generator:-

Two types of DC generator:-

- | | |
|-------------------------------------|---|
| (i) Separately excited DC generator | (ii) Self Excited DC generator |
| | <div style="display: flex; justify-content: space-around;"> <div style="text-align: center;">↓
series generator</div> <div style="text-align: center;">↓
shunt generator</div> <div style="text-align: center;">↓
compound generator</div> </div> |

* separately excited

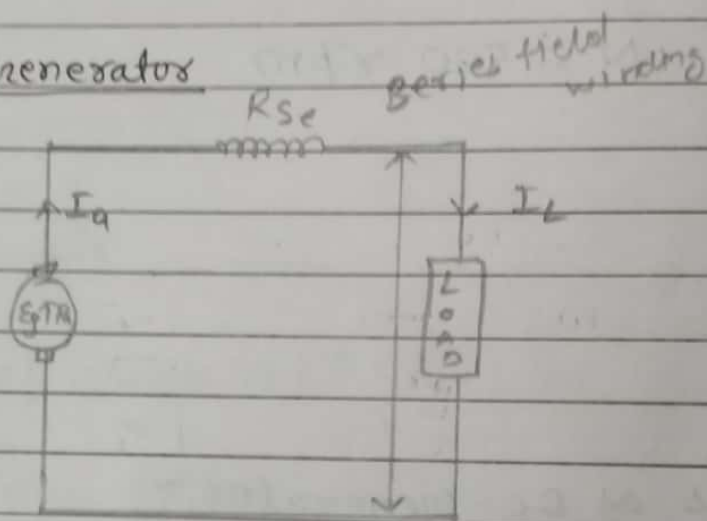


Here $I_a = I_2$ in armature
 Power developed = $E_g I_a$
 Power delivered to load = $V I_2$

where, $E_g = V + I_a R_a$

(ii) Self Excited

(a) series generator



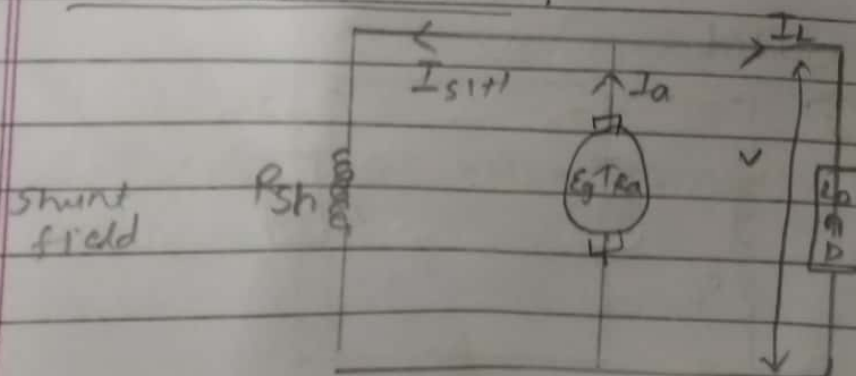
Here, $I_a = I_L$

Power developed in Armature = $E_g I_a$

Power delivered to load = $V I_L$

where $E_g = V + I_a (R_a + R_{se})$

(b) Shunt Generator:-



Here $I_a = I_L + I_{sh}$

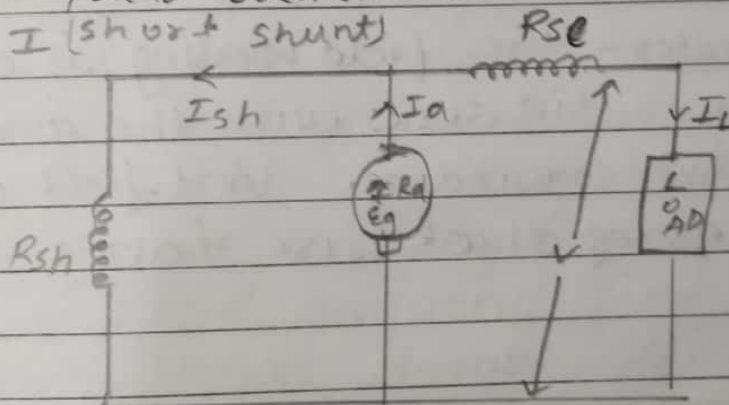
Power developed = $E_g I_a$

Power delivered to load = $V I_L$

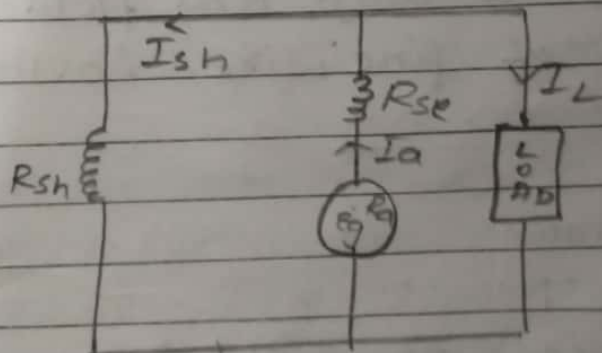
Where, $E_g = V + I_a R_a$
 $= V + (I_L + I_{sh}) R_a$

Where, $I_{sh} = \frac{V}{R_{sh}}$

(I) Compound generator
 I (short shunt)



(II) Long shunt



* separately excited generator :-

A DC generator whose field winding is supplied from an independent external DC source.
Ex - Battery

* Self excited

A DC generator whose field winding is supplied current from the output of the generator

It is further divided into three

(i) series generator - Here field winding is connected in series with the armature winding. And field winding has few turns of thick wire having low resistance

(ii) Shunt generator

When field winding is connected in parallel with the armature winding. And field winding has many turns of fine wire having high resistance.

(iii) compound generator

There are two sets of field winding on each pole, one is in series and other is in parallel with armature.

Short shunt: in which only shunt field winding is in parallel with the armature winding.

Long shunt: In which shunt field winding is in parallel both series field winding and armature winding.

DC Motor

construction is similar to dc generator. In this case we give supply to the field winding as a result constant flux produces as well as we energise the armature winding through brushes and commutator. Therefore, when a current carrying armature conductors placed in a magnetic field conductors experience force. Hence, torque develop And armature starts rotating. And we come to know electrical energy is converted into mechanical energy (rotation of armature).

Back emf - Significance of back emf (E_b):-

When the armature of DC motor rotates under the influence of driving torque armature conductors move through the magnetic field. Hence, flux is cut by the armature conductor and emf is induced in them. This induced emf always acts in opposite direction to the applied voltage.

E_b

It regulates the flow of armature current.

Armature torque of DC motor :-

$$T_a = \frac{0.159 P \phi Z I_a}{A}$$

$$T_a = \frac{0.159 P Z I_a \phi}{A}$$

where, $\frac{0.159 P Z}{A} = \text{constant}$

$$T_a \propto \phi I_a$$

Speed of DC motor

$$E_b = V - I_a R_a$$

$$\frac{P \phi Z N}{60 A} = V - I_a R_a$$

$$N = \frac{(V - I_a R_a) 60 A}{P \phi Z}$$

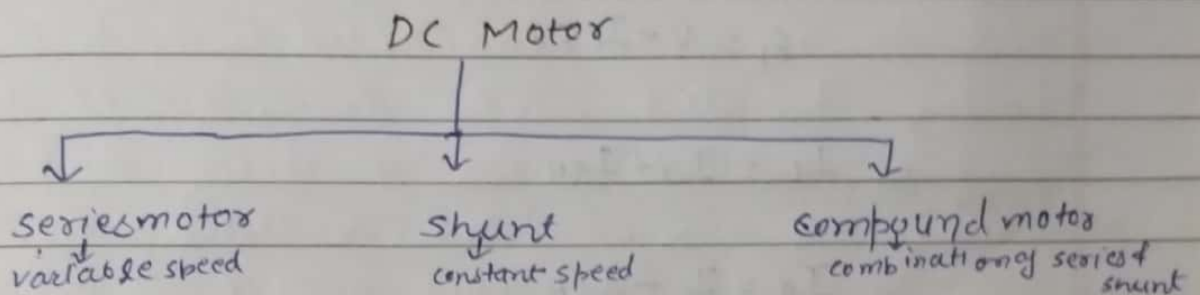
$$N = \frac{(V - I_a R_a)}{P \phi} \frac{60 A}{P Z}$$

where, $\frac{60 A}{P Z} = \text{constant values} = K$

$$N = K \cdot \frac{E_b}{\phi}$$

$N \propto \frac{E_b}{\phi}$

Types of DC Motor :-



Will & The circuit of combination of the motor will be same as the DC generator but the will be E_b instead of E_g and Reverse the flow of current only.

Application of different types of Motor.

Series motor It is a variable speed motor
ex- trains, elevator & vacuum cleaner
Hair drier.

Shunt motor — It is a constant speed motor
ex- lathe, drill, boring mills etc.

Compound motor :- It is compound motor is used
where constant speed with irregular load
or suddenly applied heavy loads.

Expresses.

Q. ~~200~~ 230 volt DC shunt motor takes a current of 40 Amp and runs at 1100 rpm. $R_a = 0.25$, $R_{sh} = 230 \Omega$, Find the emf developed by the armature.

$$I_L = 40 \text{ Amp}$$

$$R_a = 0.25 \Omega \quad R_{sh} = 230 \Omega$$

$$E_b = V - I_a R_a$$

$$I_L = I_a + I_{sh}$$

$$I_a = I_L - I_{sh}$$

$$= 40 - 1$$

$$= 39 \text{ Amp}$$

$$I_{sh} = \frac{V}{R_{sh}} = \frac{230}{230} = 1 \text{ Amp.}$$

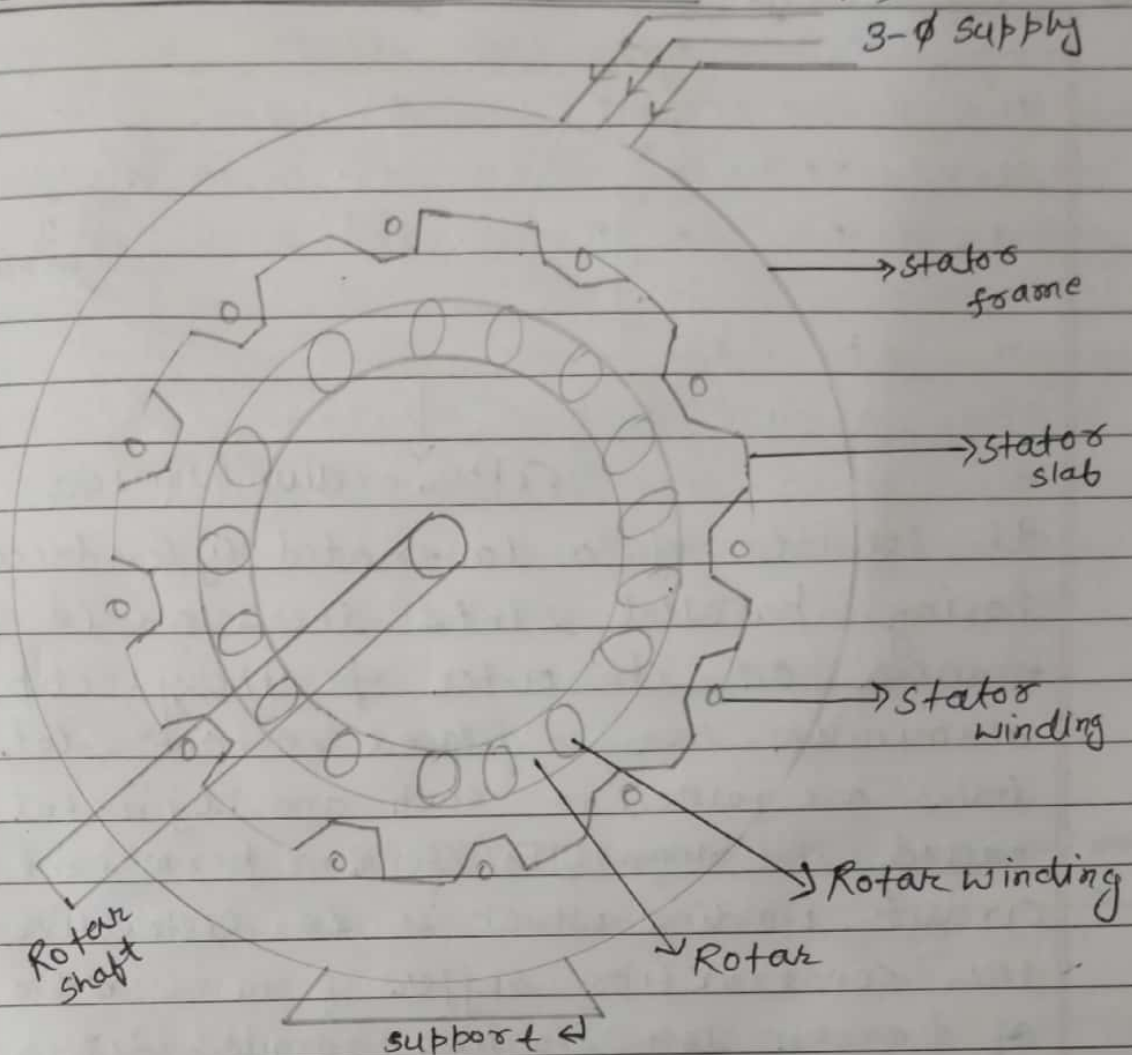
$$E_b = V - I_a R_a$$

$$= 230 - 39 \times 0.25$$

$$= 220.25 \text{ Volt} \quad \underline{\underline{\text{Ans}}}$$

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3-phase induction Motor (3- ϕ)



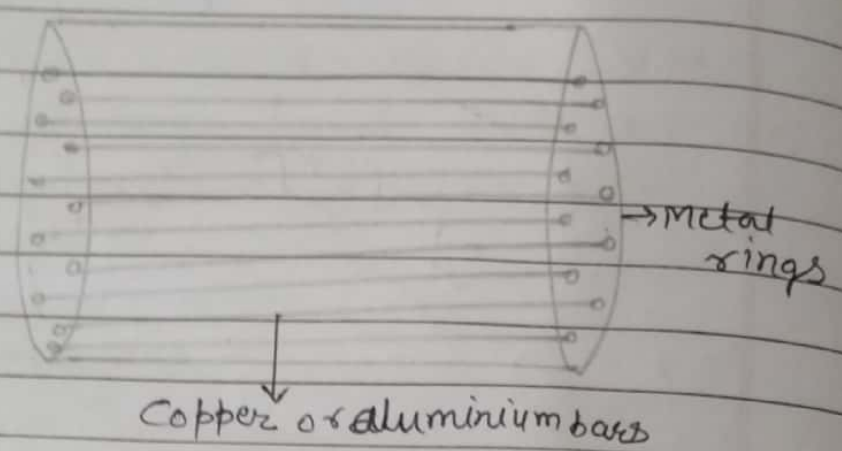
Stator - It is made up of thin lamination of silicon steel to reduce hysteresis and eddy current loss. No. of slots are provided on

conductors are placed in the stator slots and are connected to form either star or delta connected circuit.

stator frame made of cast steel.

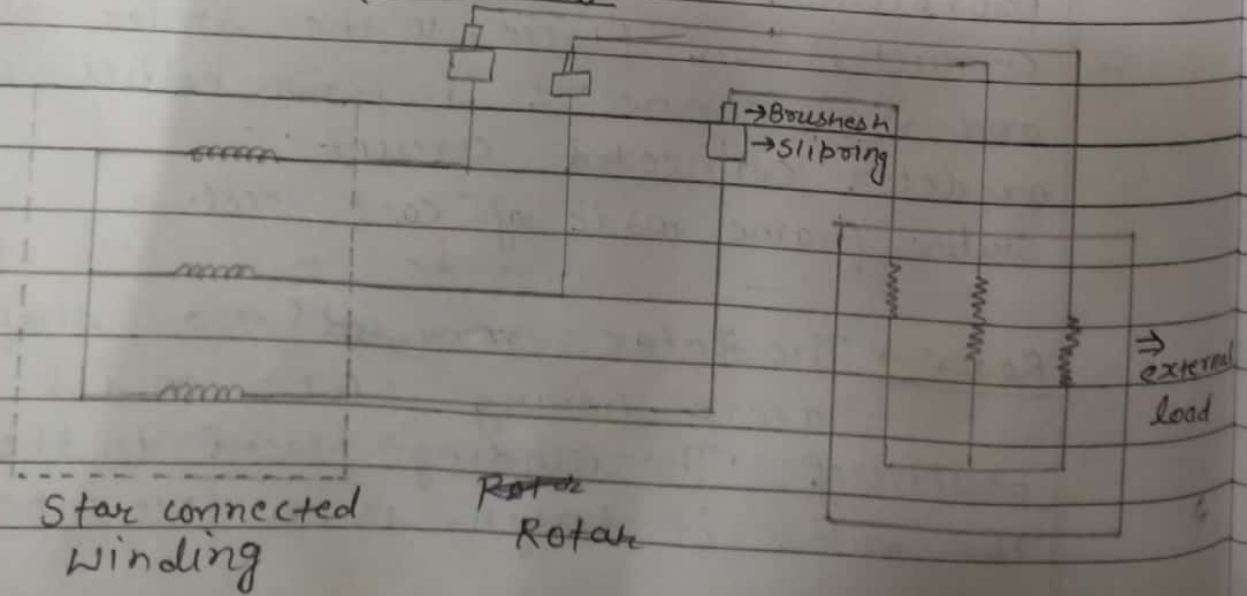
Rotor → The Rotor mounted on a shaft is a core having slots on its outer periphery. The winding placed in these slots. The rotor design is of two types:-

(i) Squirrel cage rotor design



It consists of a laminated cylindrical core having parallel slots there are in projected manner on its outer periphery. Copper and aluminium bar is placed in each slots. All these bars are joined at each end by metal rings called end rings. This form a permanent by short circuit winding which is in destructive. But this construction suffers from disadvantages of having low starting torque. differs. It is not possible to act any external resistance to the rotor circuit to get high starting torque.

(ii) Phase-wound / Slip ring



The rotor winding distributed in the slot which is usually star connected. The open ends of rotor winding are brought out and join to three insulated slip rings mounted on the rotor shaft with one brush resting on each slip rings. The three brushes are connected to three phase star connected rheostat. At starting external resistance is added in the rotor circuit to give large starting torque and these resistance are gradually decreases to zero as the motor picks up the

Synchronous speed -

The speed of revolving flux is known as synchronous speed.

$$N_s = \frac{120f}{P}$$

$$\% \text{ slip} = \frac{N_s - N_r}{N_s} \times 100$$

$$N_r = \text{Rotor speed / speed of rotor}$$

where - f = line voltage frequency in Hz
 P - number of poles

N_s - synchronous speed in rpm

Working :- • When the stator winding is connected to a balanced three phase supply, a rotating magnetic field (RMF) is setup which rotates around the stator at synchronous speed (N_s). where - $N_s = \frac{120f}{P}$.

- The RMF passes through air gap and cuts the rotor conductors, which are stationary

at start. Due to relative motion b/w RMF and the stationary rotor, an EMF is induced in the rotor conductors. Since the rotor circuit is short circuited, a current starts flowing in the rotor conductors.

- Now the current carrying rotor conductors are in a magnetic field created by the stator. As a result of this mechanical force acts on the rotor conductors. The sum of mechanical forces on all the rotor conductors produces a torque which tries to move the rotor in the same direction as the RMF.
- Hence the induction motor starts to rotate. From the above discussion, it can be seen that the three phase induction motor is self-starting motor.
- The three phase induction motor accelerates till the speed reached to a speed just below the synchronous speed.

Advantages :-

- (i) It is simple in construction.
- (ii) Little maintenance required.
- (iii) It is relatively cheaper than DC Motor.
- (iv) It has high efficiency and good power factor.
- (v) It is self started torque.

Dis Advantages:-

- It is constant speed motor. And its speed cannot be change easily.
- its starting torque is

$$\text{Torque } (T_s) \propto E_2 I_2 \cos \phi_2$$

$$T_s = K E_2 I_2 \cos \phi_2$$

$$E_2 = \text{Emf of rotor}$$

$$I_2 = \text{Current in rotor}$$

$$\cos \phi_2 = \text{rotor power factor.}$$

$$K = \text{constant.}$$

Numerical :-

Three phase induction motor is wound for 4 poles and it's supplied from 50Hz system. Calculate synchronous speed (ii) speed of motor when slip is 4%. (iii) Rotor current frequency when the motor runs at 600rpm
solⁿ-

rotor current frequency

$$f' = s f$$

where,

f' = rotor current frequency

s = slip

f = supply frequency

(i) $P = 4$
 $f = 50$

$$N_s = \frac{30}{4} \times 50$$

$$N_s = 1500 \text{ rpm}$$

(ii) Speed of motor when slip is 4%

$$\% \text{ Slip} = \frac{N_s - N_r}{N_s} \times 100$$

$$4 = \frac{1500 - N_r}{1500} \times 100$$

$$60 = 1500 - N_r$$

$$N_r = 1500 - 60$$

$$N_r = 1440 \text{ rpm}$$

(iii) $N_r = 600 \text{ rpm}$

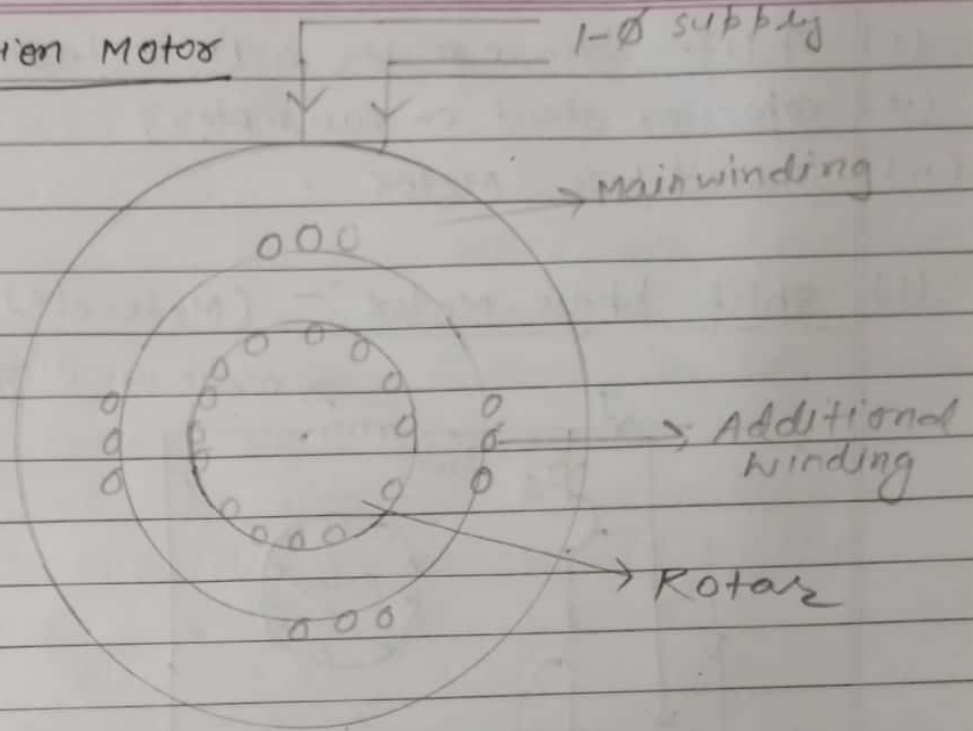
Rotor current frequency = ?

$$\text{Slip} = \frac{1500 - 600}{1500} = \frac{900}{1500} = \frac{3}{5}$$

$$f' = sf$$

$$f' = \frac{3}{5} \times 50 = 30 \text{ Hz} \quad \text{Ans.}$$

1- ϕ induction Motor



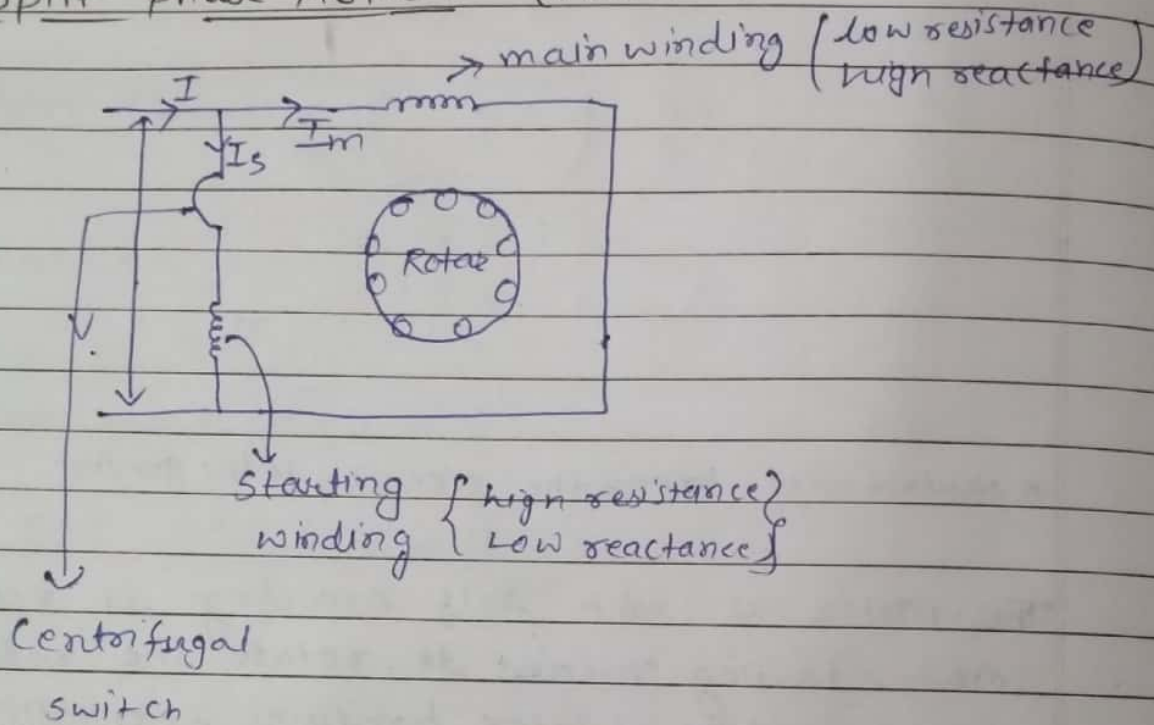
* Neutral wire brings the circuit to the ground.

This motor is not self starting it requires some starting means to rotate the rotor because the stator produces alternating magnetic field. and the two components of these flux cancel each other hence no net torque develops as to rotate the rotor rotating magnetic field is required. To make the motor self starting we should somehow produce a revolving stator magnetic field which can be achieved by converting single phase supply into two phase supply by additional winding when the motor attains sufficient speed (about 75% of synchronous speed) then starting winding may be removed depend on type of motor.

Types of 1- ϕ induction Motor

- (i) split-phase motor, (ii) capacitor start motor
- (iii) capacitor start c-run motor
- (iv) shaded pole motor

III Split phase motor :- (Moderate)



$$T = K I_s I_m \sin \alpha$$

where

I_s = starting current

I_m = Main current

α = phase diff b/w I_s & I_m

$\alpha = 25^\circ \text{ to } 30^\circ$

Stator of split phase 1- ϕ induction motor provided ~~added~~ with starting winding in addition to main winding. starting winding is located 90° from main winding. which operates only when the motor starts. starting winding ~~has~~ has high resistance and low reactance whereas main winding has low resistance and high

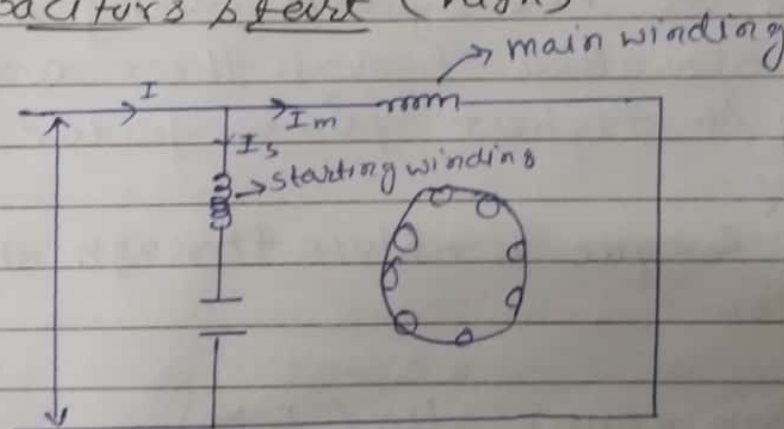
reactance. There is small phase diff α (25° to 30°) b/w I_s & I_m .

When the motor is converted into two phase winding only then we can achieve rotational magnetic field with the help of which rotor starts rotating. Once the rotor starts rotating we can disconnect starting winding from the main winding by opening this centrifugal switch.

$$T_s = K I_s I_m \sin \alpha$$

Ex - washing machine, fan small machine

(ii) Capacitors Start (high)

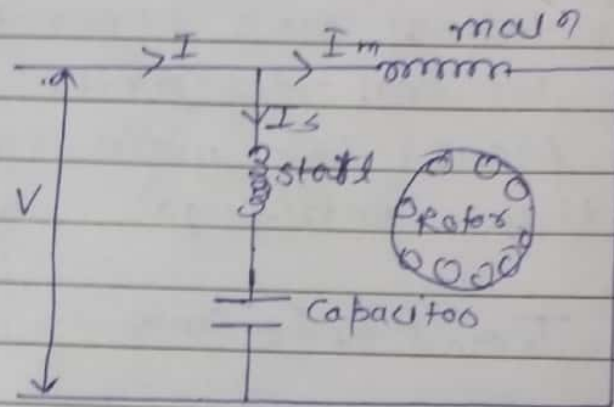


In these construction we will connect capacitor in series with starting winding the use of these capacitor to make a phase angle α approaches around 90° b/w I_m & I_s these help to increase the starting torque when the rotor maintain the speed of around 75% of synchronous speed these starting winding

opening centrifugal switch. After that motor attain certain speed by ϕ phase, we can achieve high torque.

Ex - large fans

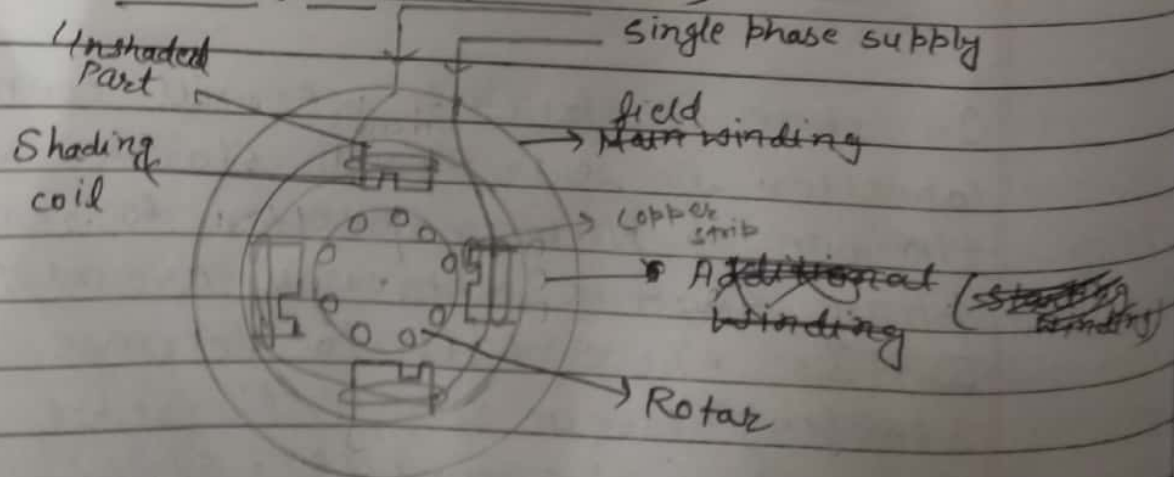
(iii) CSCR (highest)



In these combination capacitor will remain connected to the system during starting and running operation through these combination we can get highest power factor and higher efficiency.

- constant torque is obtain through there combination.

(iv) Shaded pole region :- (1- ϕ I.M)



Construction -

These type of motor have salient pole on this stator and squirrel cage design of rotor. The poles has slot cut at about one third distance from one edge, where copper strip is placed called shading coil. This part is considered shaded part and rest of the part is considered unshaded part.

Working:- When alternating current pass through field winding surrounded about whole pole axis of pole shifts from unshaded part to shaded part. This shifting of magnetic axis is equivalent to actual physical movement of the pole. Hence the rotor will move in the same direction as the pole.

These motor have low starting torque, low efficiency, low power factor.

These motor only suitable for low power consumption. ex - toys, very small fans etc.

V.V.
Imp
#

Emf eqn of alternator (synchronous generator)

In one revolution each stator conductor cut by magnetic flux.

change in flux -

$$d\phi = p\phi$$

time take to complete one revolution.

$$dt = \frac{60}{N}$$

Emf induced in each stator conductor = $\frac{d\phi}{dt}$

$$= \frac{P\phi}{\frac{60}{N}} = \frac{P\phi N}{60}$$

Suppose there are Z conductors in series per phase

$$\text{Average emf/phase} = \frac{P\phi Z N}{60}$$

$$\text{As, } N = \frac{120f}{P}$$

Average emf per phase

$$E_{ph} = \frac{P\phi Z \cdot 120f}{P \times 60}$$

$$E_{ph} = 2\phi Z f$$

R.M.S emf per phase = $2f\phi Z \times 1.11$ (form factor)

$$E_{ph} = 2.22 f \phi Z$$

$$E_{ph} = 2.22 f \phi Z \times K_p \times K_d$$

$$= 2.22 f \phi Z \times K_w$$

$$\because K_w = K_p \times K_d$$

Star connection = $V_L = \sqrt{3} V_{ph}$

Delta connection = $I_L = \sqrt{3} I_{ph}$

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$$E_L = \sqrt{3} E_{ph}$$

Numerical :-

A. Find the number of Armature conductors in series per phase required for the armature of 3 phase, 50 Hz, 10 pole Alternator. The winding is star connected to give a line voltage 11000 volts. The flux per pole is 0.16 weber. assume $k_p = 1$ & $k_d = 0.96$

Solⁿ—

↓ Pitch factor
↓ Distribution factor

$$K_w = k_p \times k_d$$

↓ winding factor

Given

$P = 10$

$\phi = 0.16$ weber

$f = 50$ Hz

$E_L = 11000$

$Z = ?$

$$E_{ph} = 2.22 f \phi Z k_p k_d$$

$$\frac{11000}{\sqrt{3}} = 2.22 \times 50 \times 0.16 \times Z \times 1 \times 0.96$$

$$Z = \frac{11000}{1.732 \times 2.22 \times 50 \times 0.16 \times 0.96}$$

$Z = 372.50$

$Z = 372$ Ans

5. Electrical Installations

18/11/23

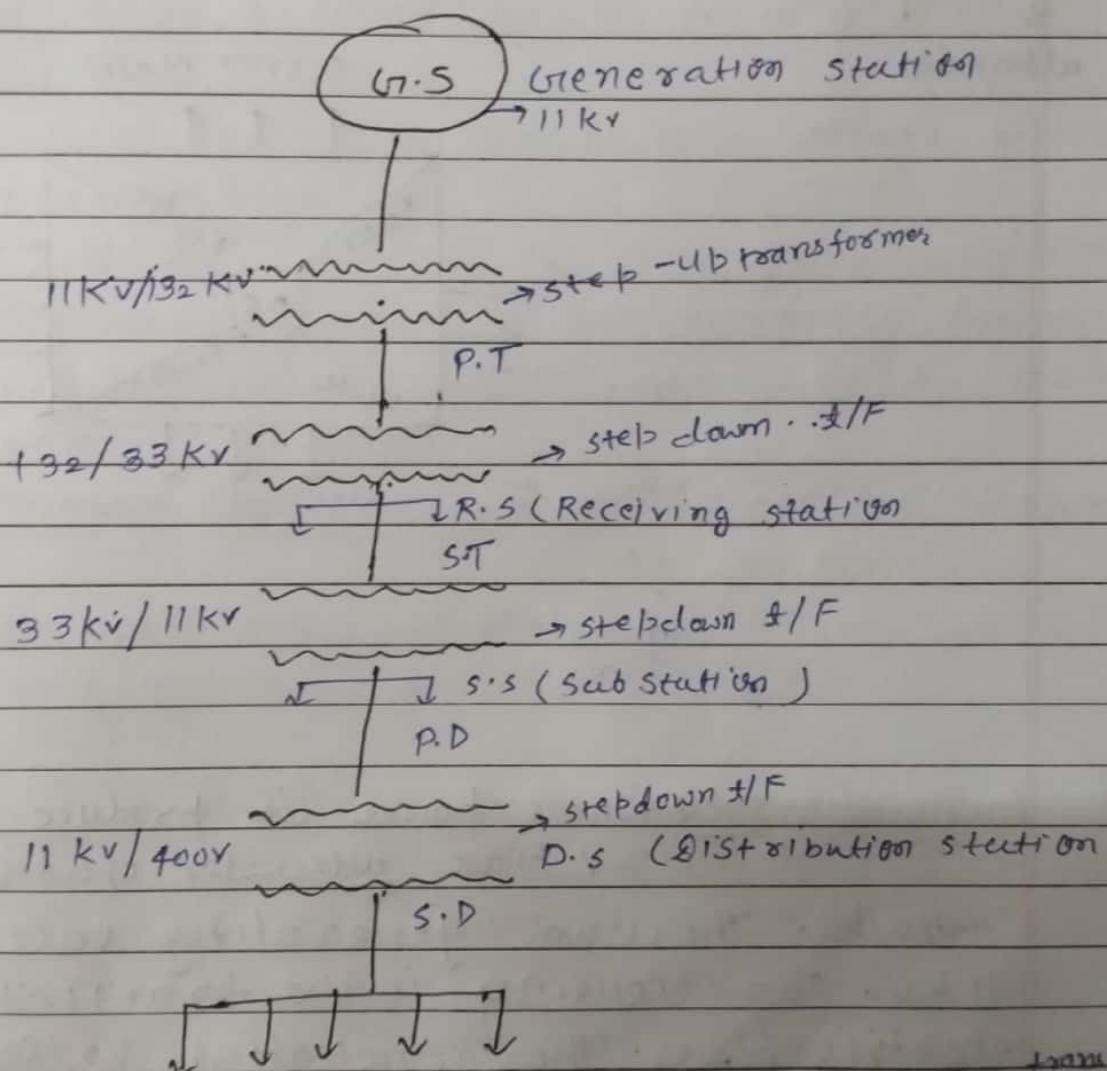
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Layout of power supply / power system

Power system :- The system consists of generation, transmission and distribution

Generate $\rightarrow$ Transmission $\rightarrow$ Distribution.

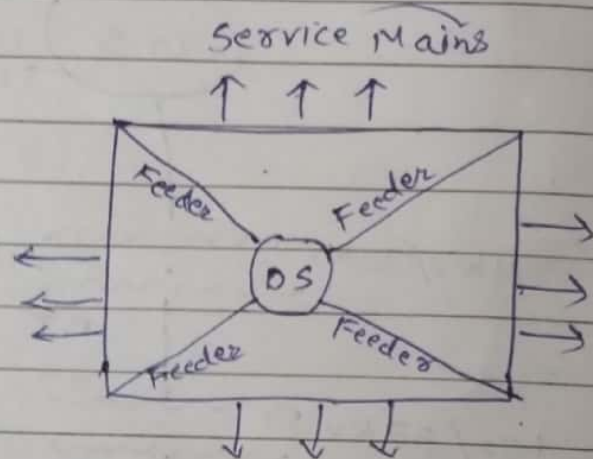
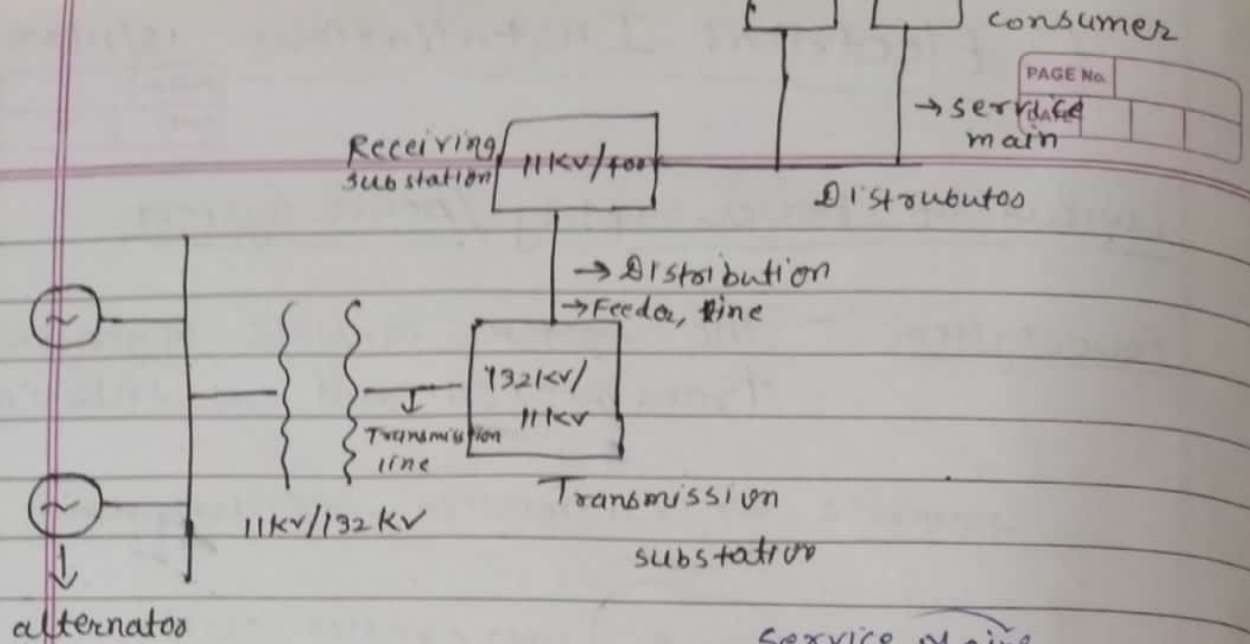


P.T = Primary transformer

S.T = Secondary "

P.D = Primary distribution

S.D = Secondary distribution



Generating station:- Where power is produced from 3- ϕ phase alternator operating in parallel. The usual generating voltage is 11 kv. For economy in the transmission of electric power. The generating voltage is stepped up to 132kv at the generating station with the help of 3-phase step up transformer.

1. The transmission of electric power at high voltage has following advantages —

- (i) saving of conductor material due to ~~the~~ decrease in current
- (ii) For economy purpose.
- (iii) High transmission efficiency.

But there is some limit. to which can be increase.

Because increase in transmission voltage ^{introduced} insulation problem.

- (iv) As well as ^{the} cost of switch gear and transformer ~~eq~~ equipment. generally primary transmission is carried out at 66KV, 132KV, 220KV and 400KV.

~~(i)~~ primary transmission - The electric power at 132KV is transmitted by 3-phase 3 wire over head system to out skirts.

(ii) secondary transmission :- The primary transmission terminates at receiving station. at the receiving station the voltage is further reduce to 33KV by step down transformer. From this station electric power is transmitted at 33KV by 3-phase 3 wire over head system to various substation. located at the some points in the city.

Primary Distribution. - The secondary transmission line terminates at the substation where voltage is reduced from 33 kV to 11 kV which is generally supply for ^{big} consumers (big industries)

Secondary distribution - The electric power from Primary distribution 11 kV is delivered to distribution sub station. These sub station are located near the consumer locality where voltage is

stepdown to 400V. The voltage b/w any two phases 400 and b/w any phase and neutral is 230V.

secondary 3- ϕ phase four wire system is used for secondary Distribution.

Feeder :- It is a conductor which connects the substation to the area where power is to be distributed.

No tappings are taken from the feeder. so that current remains the same.

Distributor - It is a conductor from which tappings are taken for supply to the consumer. AB, BC, CD, DA is a distributor.

Service main - It is small cable which connects the distributor to the consumer terminals.

Transmission voltage :-

$$\text{Transmission Voltage} = 5.5 \sqrt{\frac{0.6217 \times 3P}{150}} \text{ Kv}$$

V = Voltage in Kv

L = length in Kw

P = power to be distributed (Kw)

Switch gear :- ^{Device} A operators, used for switching, controlling, ~~feet~~ protecting the electrical circuit and electrical equipment is called switch gear. ex-switch, fuse, MCB, circuit breaker.

relays etc.
↳ sensing

Switch fuse unit

Switch - switch is a device used for opening or closed and electric circuit ^{only} but it cannot interrupt the current.

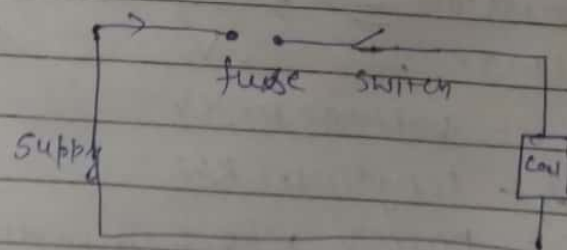
Fuse :- A fuse is a short piece of wire which melts when excessive current flows through it for sufficient time. It is always inserted in series with the circuit to be protected. Hence fuse perform the function of detection of high current as well as interruption function of the circuit. ~~fault current~~

operating time - 0.02 sec

O.P. time 0.02

Advantages - Cheap, No maintenance, break circuit without any noise, ~~high~~ high resistance and low melting point.

Disadvantages - It has to be removed after each operation.



Circuit Breaker :-

It consists of fixed and moving contact. It is an ~~equi~~ device which can open or closed.

a circuit under No load ~~to~~ full load at fault condition.

It can be operate manually as well as automatically under fault condition. operating time - 0.1 to 0.02 sec

* No replacement after operation.

Grid :- It is a network which interconnects the Generation transmission, and distribution unit

Advantages : ① It make the system economical and reliable

② If there is sudden increase of load & loss of generation of power in any zone then it borrow from the adjacent inter connected area.

Two types of Grid

(i) Regional Grid - It is form by inter connecting different transmission system of a particular area to the through the transmission line

(ii) National Grid - by inter connecting regional grid.

Numerical :- A toaster consumes 1200 Watt power in 30 minutes calculate the energy used by the toaster in the month of June.

Solⁿ-

$$P = 1200 \text{ watt}$$

$$T = 30 \text{ min}$$

$$\text{Energy used} = P \times T$$

$$\text{Energy used} = \left(\frac{1200}{1000} \text{ kW} \right) \times \left(\frac{30}{60} \right) \text{ h}$$

$$= \frac{1200}{1000} \times \frac{30}{60}$$

$$= 0.6$$

$$\text{Energy used - per day} = 0.6 \text{ Kwh}$$

$$\text{Energy used in 30 days} = 30 \times 0.6$$

$$= 18 \text{ Kwh}$$